



I'm Arjun Srinivasan

- Lead Research Scientist at Tableau (Salesforce) in Seattle, WA

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Natural language and search systems



Dashboard authoring & consumption

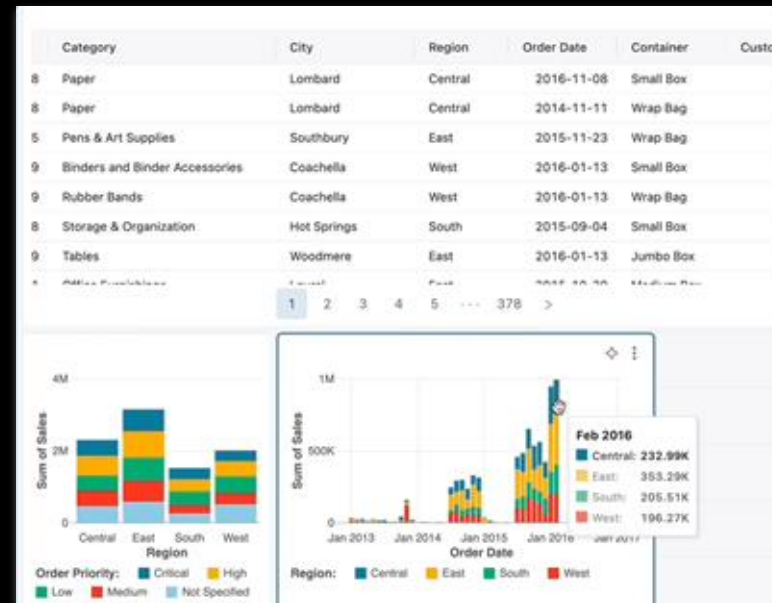
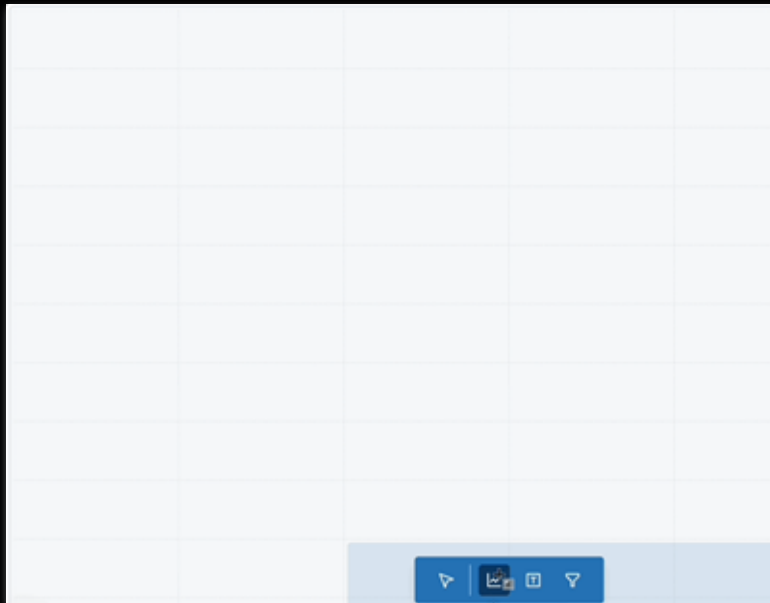


Data-driven communication



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- Lead Research Scientist at Tableau (Salesforce) in Seattle, WA
- Previous:
 - Software Engineer at Databricks (AI/BI Dashboards)



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- Lead Research Scientist at Tableau (Salesforce) in Seattle, WA
- Previous:
 - Software Engineer at Databricks (AI/BI Dashboards)
 - CS + HCI PhD at Georgia Tech

Thesis:
Combining Natural Language and Direct Manipulation for Human-Data
Interaction through Visualizations

🏆 IEEE VGTC Dissertation Award



Interaction in Information Visualization

Arjun Srinivasan

IDC School of Design | Sep 2026

Agenda

- What is interaction and why should we think about it in the context of information visualization?
- Review categories of visualization interactions
- (Briefly) discuss post-WIMP interaction with visualizations

Takeaways

- An understanding of different types of interactions supported by visualizations + examples of analytic experiences interactions can afford
- Design considerations when working on visualization systems
- Strategies and language for prompting ChatGPT/Claude and testing results
- An appreciation for (/interest in) visualization research

“The effectiveness of information visualization hinges on two things: its ability to clearly and accurately represent information and our ability to interact with it to figure out what the information means.”

S. Few

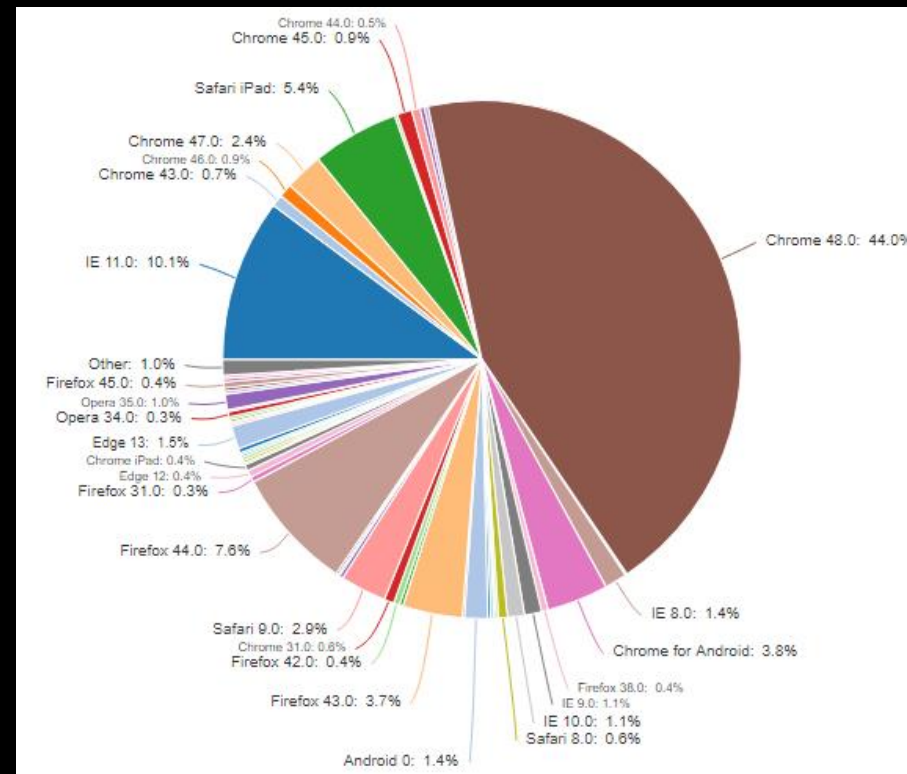
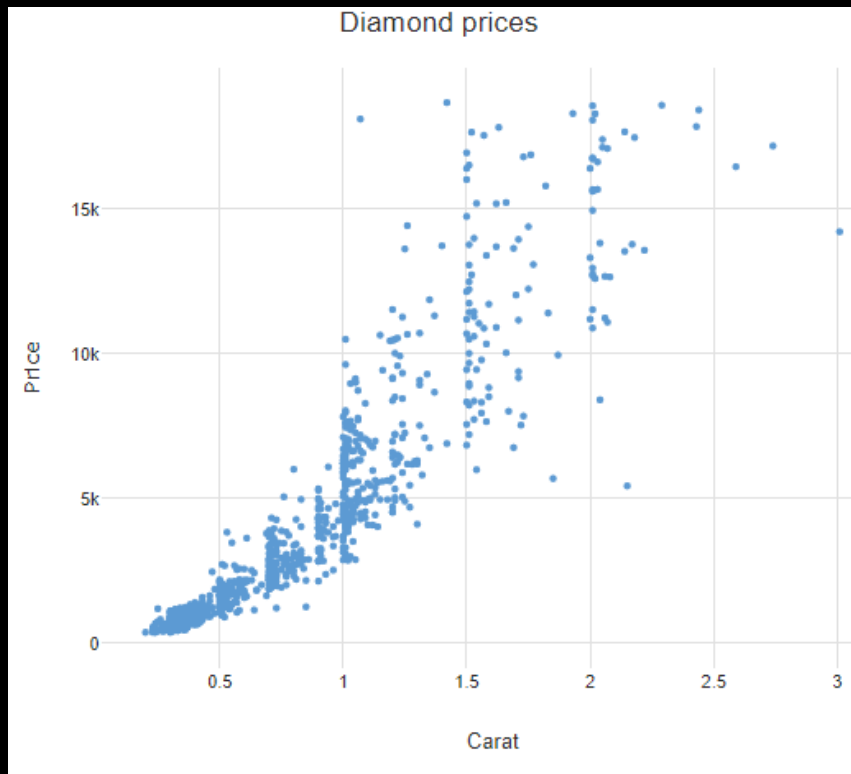
Now You See It, p. 55

Why interaction?



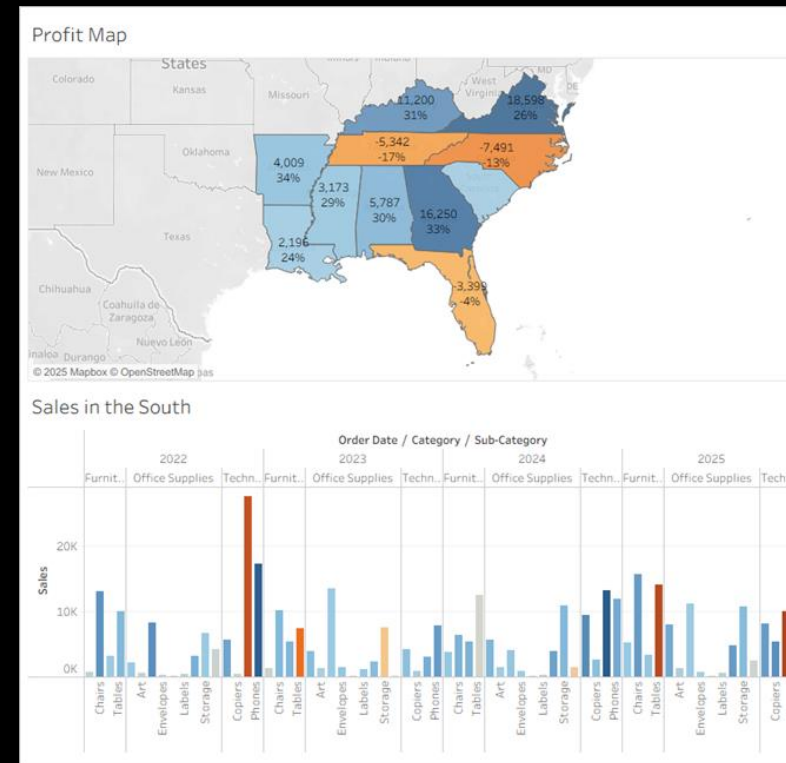
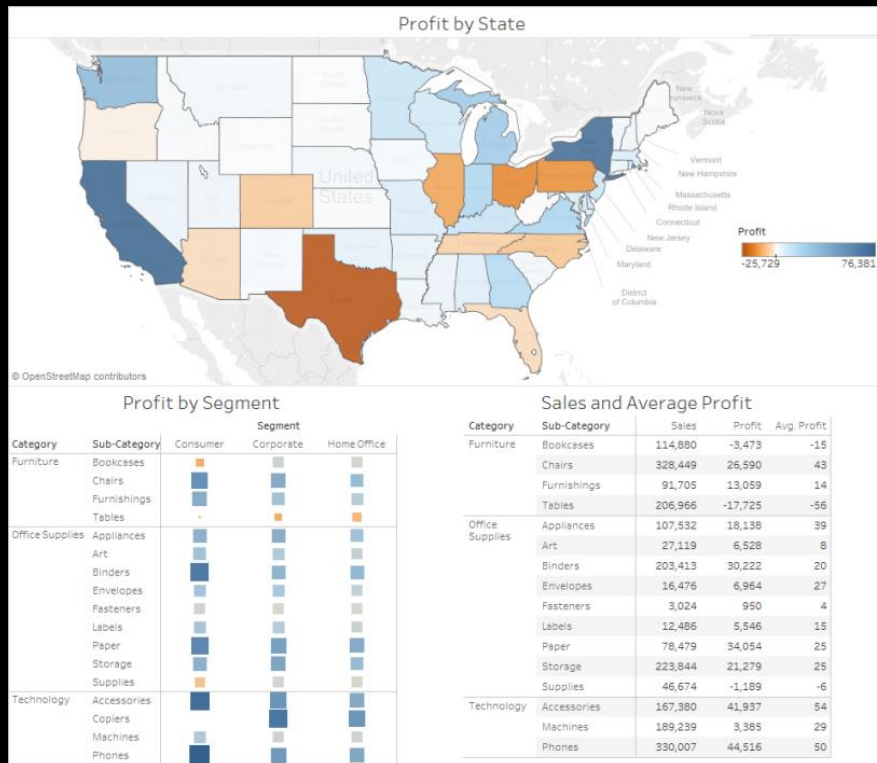
Why interaction?

- Representations can be dense



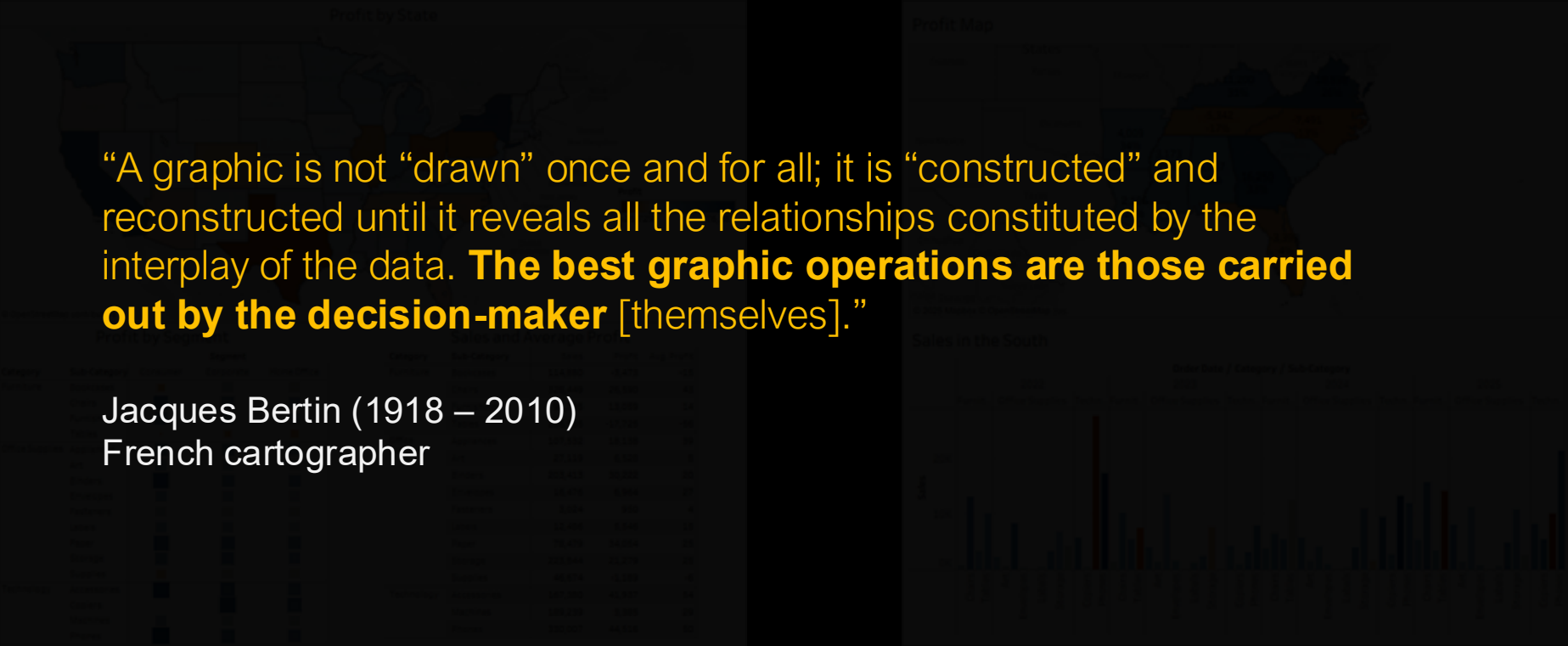
Why interaction?

- Static charts display a specific instance of the data or pre-determined perspectives



Why interaction?

- Static charts display a specific instance of the data or pre-determined perspectives

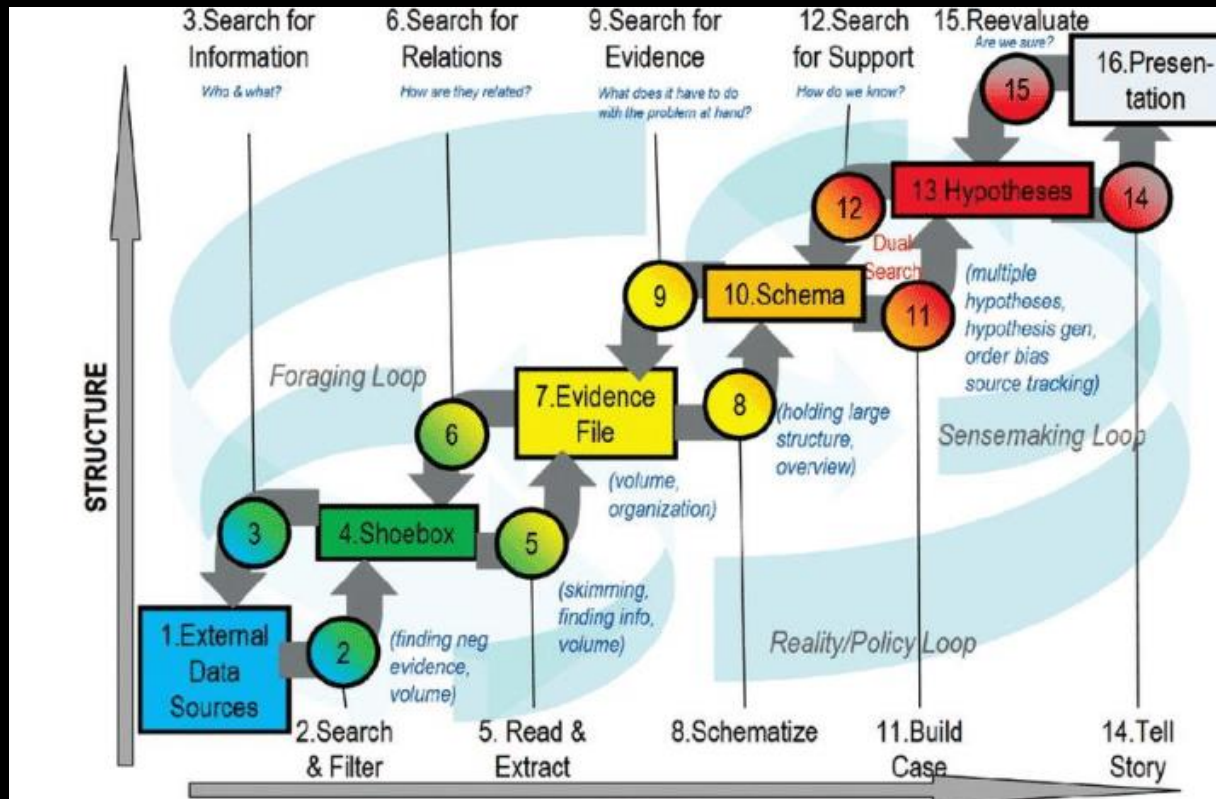


“A graphic is not “drawn” once and for all; it is “constructed” and reconstructed until it reveals all the relationships constituted by the interplay of the data. **The best graphic operations are those carried out by the decision-maker** [themselves].”

Jacques Bertin (1918 – 2010)
French cartographer

Why interaction?

- Analysis is a process



Pirolli and Card, 2005



What are the different "categories" or "types" of interactions in visualization tools?

The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations

Ben Shneiderman

Department of Computer Science,
Human-Computer Interaction Laboratory, and Institute for Systems Research
University of Maryland
College Park, Maryland 20742 USA
ben@cs.umd.edu

Abstract: A useful starting point for designing advanced graphical user interfaces is the Visual Information-Seeking Mantra: **Overview first, zoom and filter, then details-on-demand**. But this is only a starting point in trying to understand the rich and varied set of information visualizations that have been proposed in recent years. This paper offers a task by data type taxonomy with seven data types (1-, 2-, 3-dimensional data, temporal and multi-dimensional data, and tree and network data) and seven tasks (overview, zoom, filter, details-on-demand, relate, history, and extract).

understood information need (known-item search) to developing an understanding of unexpected patterns within the collection (browse) ([Marchionini, 1995](#)).

Exploring information collections becomes increasingly difficult as the volume grows. A page of information is easy to explore, but when the information becomes the size of a book, or library, or even larger, it may be difficult to locate known items or to browse to gain an overview.

Designers are just discovering how to use the rapid and high resolution color displays to present large amounts of information in orderly and user-controlled

*Overview first, **zoom & filter**, then **details on demand***

The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations

Ben Shneiderman

Starting Simple – adding value to static visualisation through simple interaction AVI98 paper

Alan Dix[†] and Geoffrey Ellis[‡]

[†] School of Computing, Staffordshire University, UK

A.J.Dix@soc.staffs.ac.uk
<http://www.soc.staffs.ac.uk/~cmtajd/>

[‡] School of Computing and Mathematics, Huddersfield University, UK

g.p.ellis@hud.ac.uk

Abstract

Interactive visualisation has been one of the most exciting areas in HCI over recent years. The key term here is 'interactive', and in this paper we assert that virtually any static representation can become more powerful by the addition of simple interactive elements. This is demonstrated by adding interactivity to standard representations including stacked histograms, pie charts and scatter plots. We show how adding interactivity can help resolve many of the trade-offs inherent in static visualisations by allowing multiple options to be available and most importantly for them to be interactively related. Many years of creativity and effort have been invested in traditional generic and bespoke visualisations. Adding interactivity leverages this accumulated experience, however it adds an extra dimension.

- Highlighting and focus
- Accessing extra info – drill down and hyperlinks
- Overview and context – zooming and fisheyes
- Same representation, changing parameters
- Linking representations – temporal fusion

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Abstract

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Information Visualization and Visual Data Mining

Daniel A. Keim

Abstract—Never before in history data has been generated at such high volumes as it is today. Exploring and analyzing the vast volumes of data becomes increasingly difficult. Information visualization and visual data mining can help to deal with the flood of information. The advantage of visual data exploration is that the user is directly involved in the data mining process. There is a large number of information visualization techniques which have been developed over the last decade to support the exploration of large data sets. In this paper, we propose a classification of information visualization and visual data mining techniques which is based on the data type to be visualized, the visualization technique and the interaction and distortion technique. We exemplify the classification using a few examples, most of them referring to techniques and systems presented in this special issue.

Keywords—Information Visualization, Visual Data Mining, Visual Data Exploration, Classification

Benefits of Visual Data Exploration

For data mining to be effective, it is important to include the human in the data exploration process and combine the flexibility, creativity, and general knowledge of the human with the enormous storage capacity and the computational power of today's computers. Visual data exploration aims at integrating the human in the data exploration process, applying its perceptual abilities to the large data sets available in today's computer systems. The basic idea of visual data exploration is to present the data in some visual form, allowing the human to get insight into the data, draw conclusions, and directly interact with the data. Visual data mining techniques have proven to be of high value in exploratory data analysis and they also have a high potential

- Projection
- Filtering
- Zooming
- Distortion
- Linking and brushing

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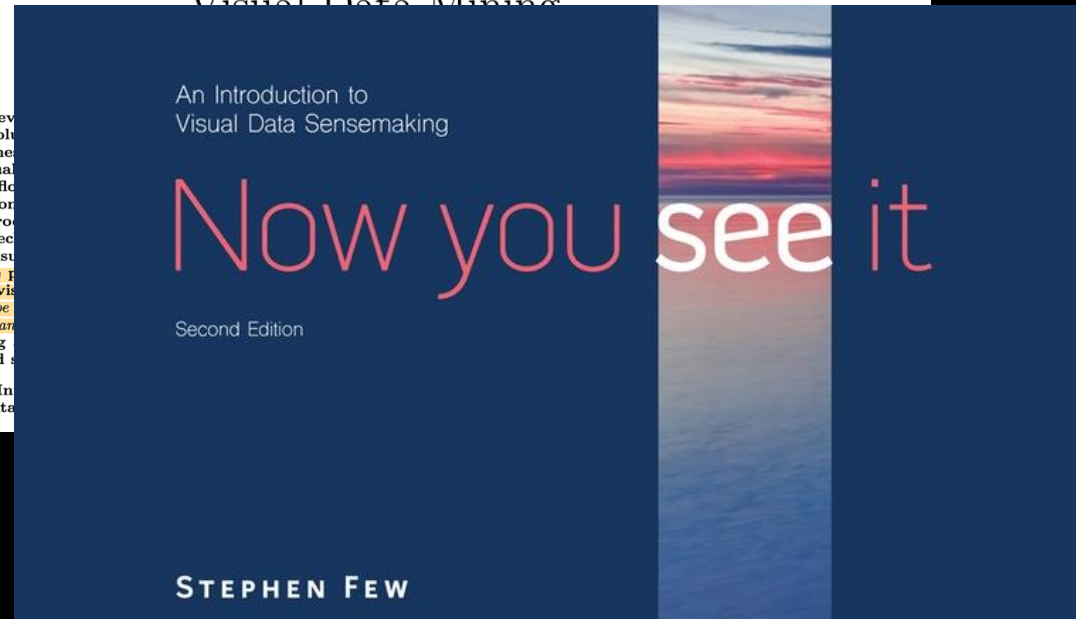
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Keywords— In
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Information Visualization and Visual Data Mining



- Comparing
- Sorting
- Adding variables
- Filtering
- Highlighting
- Aggregating
- Re-expressing
- Re-visualizing
- Zooming and panning
- Re-scaling
- Accessing details on demand
- Annotating
- Bookmarking

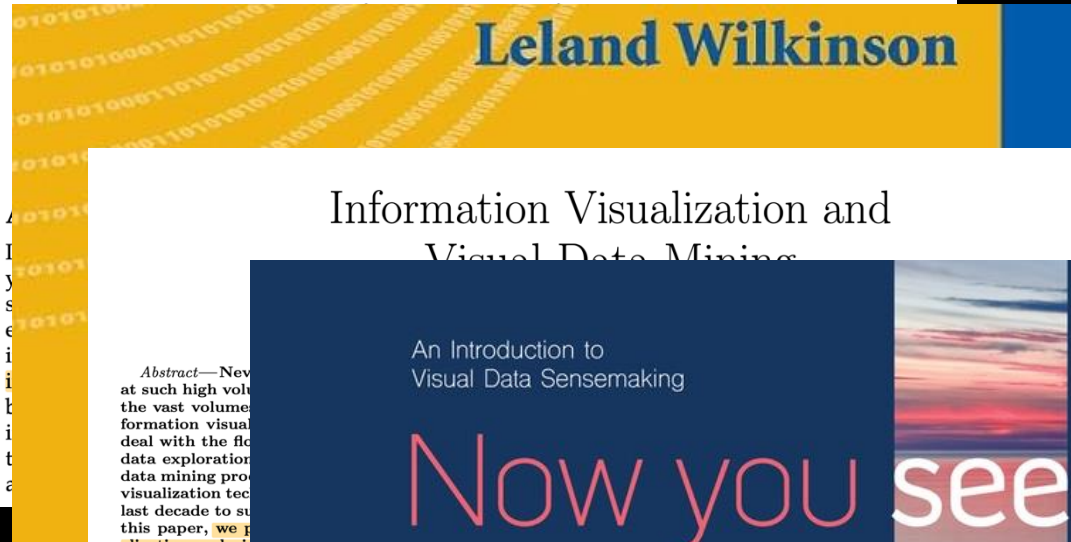
(there are many more...)

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Information Visualization and Visual Data Mining

An Introduction to
Visual Data Sensemaking

Now you see it

What is Interaction for Data Visualization?

Evanthia Dimara and Charles Perin*

Abstract—Interaction is fundamental to data visualization, but what “interaction” means in the context of visualization is ambiguous and confusing. We argue that this confusion is due to a lack of consensual definition. To tackle this problem, we start by synthesizing an *inclusive view* of interaction in the visualization community – including insights from information visualization, visual analytics and scientific visualization, as well as the input of both senior and junior visualization researchers. Once this view takes shape, we look at how interaction is defined in the field of human-computer interaction (HCI). By extracting commonalities and differences between the views of interaction in visualization and in HCI, we synthesize a *definition of interaction for visualization*. Our definition is meant to be a thinking tool and inspire novel and bolder interaction design practices. We hope that by better understanding what interaction in visualization is and what it can be, we will enrich the quality of interaction in visualization systems and empower those who use them.

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Interaction seems like a difficult thing to pin down and characterize from a visualization design and system standpoint

...What if we go back to the user trying to solve problems and take a user-centered perspective on characterizing interaction?

What is Interaction for Data Visualization?

Evanthia Dimara and Charles Perin*

Abstract—Interaction is fundamental to data visualization, but what "interaction" means in the context of visualization is ambiguous and confusing. We argue that this confusion is due to a lack of consensual definition. To tackle this problem, we start by synthesizing an *inclusive view* of interaction in the visualization community – including insights from information visualization, visual analytics and scientific visualization, as well as the input of both senior and junior visualization researchers. Once this view takes shape, we look at how interaction is defined in the field of human-computer interaction (HCI). By extracting commonalities and differences between the views of interaction in visualization and in HCI, we synthesize a *definition of interaction for visualization*. Our definition is meant to be a thinking tool and inspire novel and bolder interaction design practices. We hope that by better understanding what interaction in visualization is and what it can be, we will enrich the quality of interaction in visualization systems and empower those who use them.

Toward a Deeper Understanding of the Role of Interaction in Information Visualization

Ji Soo Yi, Youn ah Kang, John T. Stasko, *Member, IEEE*, and Julie A. Jacko

Abstract—Even though interaction is an important part of information visualization (Infovis), it has garnered a relatively low level of attention from the Infovis community. A few frameworks and taxonomies of Infovis interaction techniques exist, but they typically focus on low-level operations and do not address the variety of benefits interaction provides. After conducting an extensive review of Infovis systems and their interactive capabilities, we propose seven general categories of interaction techniques widely used in Infovis: 1) Select, 2) Explore, 3) Reconfigure, 4) Encode, 5) Abstract/Elaborate, 6) Filter, and 7) Connect. These categories are organized around a user's intent while interacting with a system rather than the low-level interaction techniques provided by a system. The categories can act as a framework to help discuss and evaluate interaction techniques and hopefully lay an initial foundation toward a deeper understanding and a science of interaction.

Index Terms—Information visualization, interaction, interaction techniques, taxonomy, visual analytics



- Survey

- 59 papers

- Papers introducing new interaction systems
 - Well-known papers in subareas of InfoVis

- 51 systems

Abstract—Even though interaction is an important part of information visualization (Infovis), it has garnered a relatively low level of academic and commercial interest. Although many commercial systems exist, but they typically focus on low-level operations and do not address the variety of benefits interaction provides. After conducting an empirical study of 51 systems, we propose seven general categories of interaction techniques widely used in Infovis: 1) Select, 2) Explore, 3) Reconfigure, 4) Encode, 5) Abstract/Elaborate, 6) Filter, and 7) Connect. These categories are organized around a user's intent while interacting with a system rather than the low-level interaction techniques provided by a system. The categories can act as a framework to help discuss and evaluate interaction techniques and hopefully lay an initial foundation toward a deeper understanding and a science of interaction.

Index Terms—Information visualization, interaction, interaction techniques, taxonomy, visual analytics

- Affinity Diagramming to find themes

Select—mark something as interesting.

Abstract/Elaborate—show me more or less detail.

Explore—show me something else.

Connect—show me related items.

Reconfigure—show me a different arrangement.

Filter—show me something conditionally.

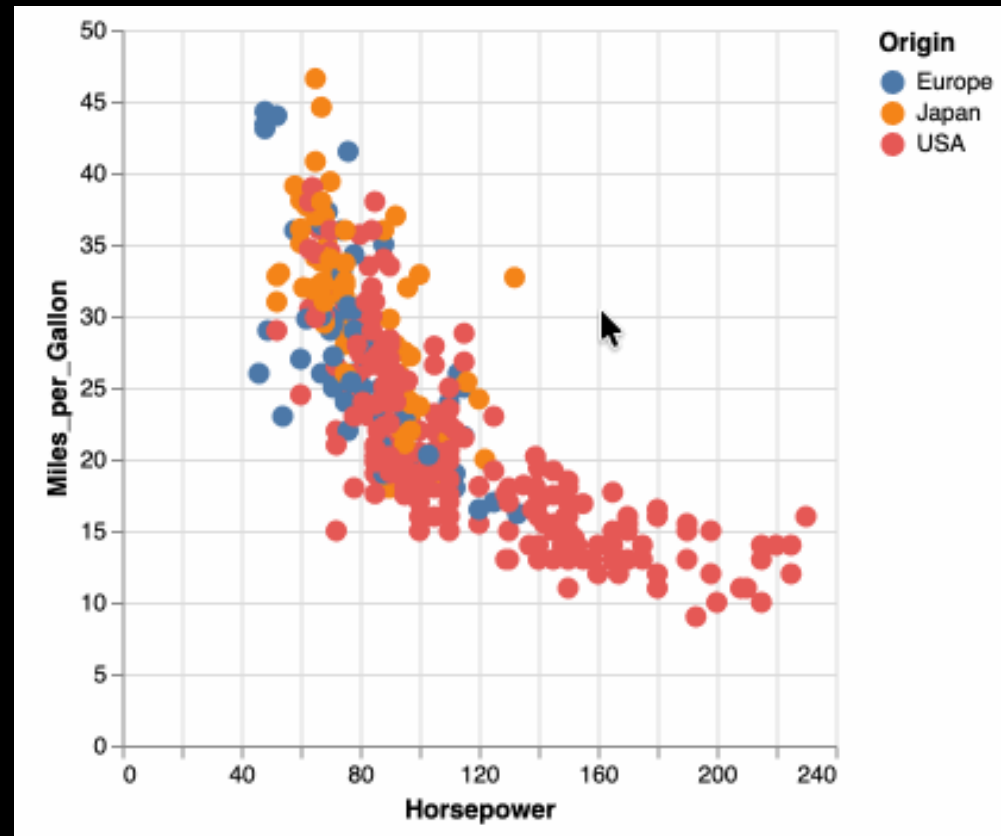
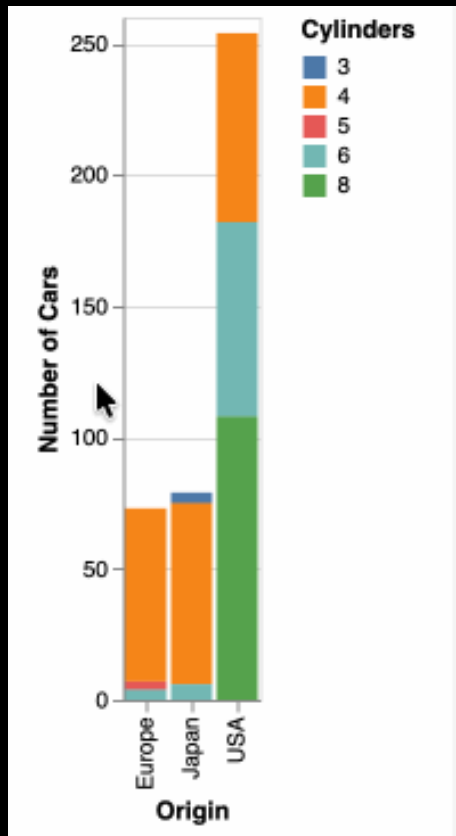
Encode—show me a different representation.

Select (“Mark something as interesting”)

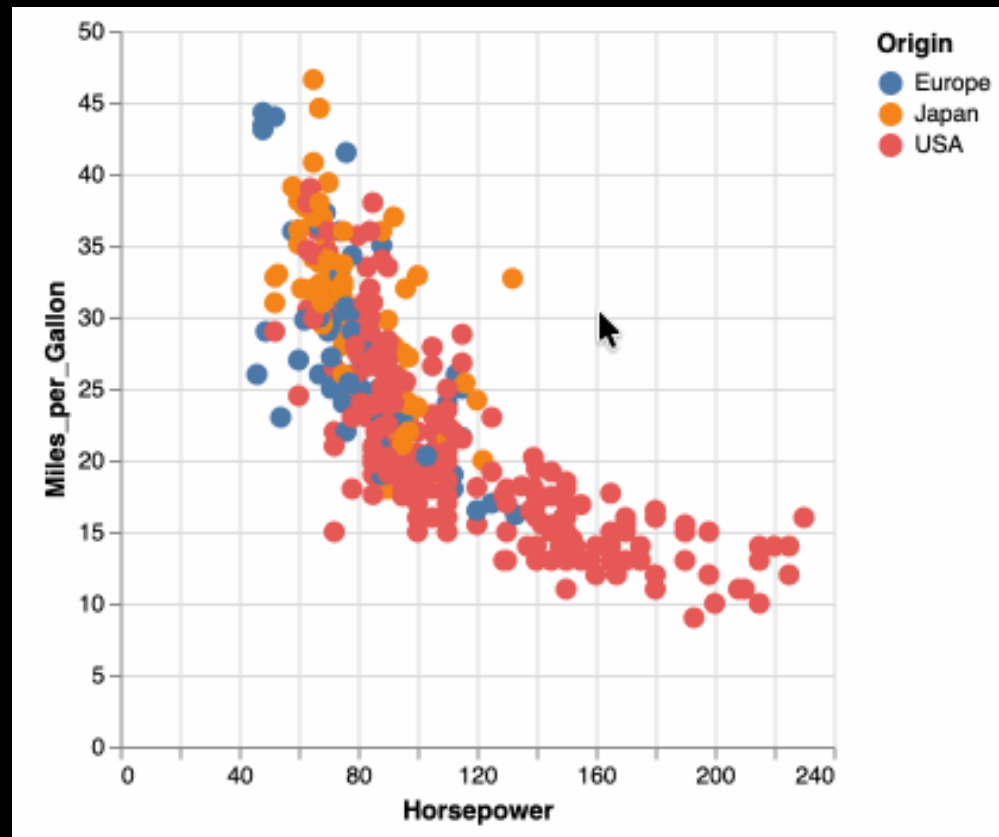
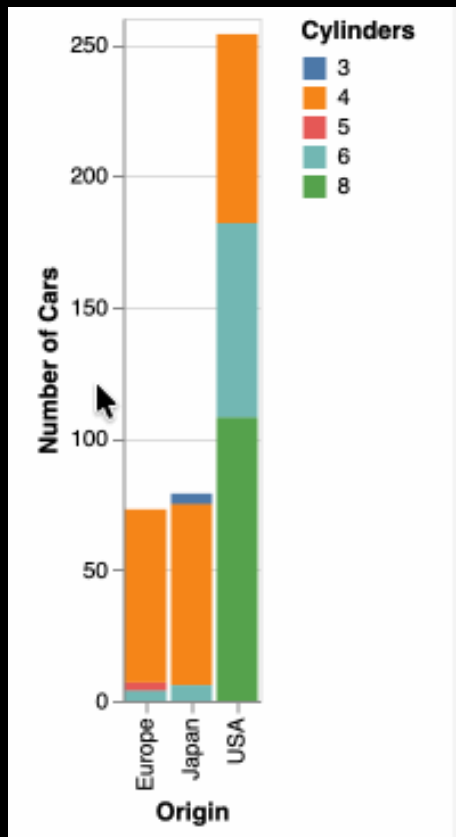
Typically supported through hover/click/lasso

Often a precursor interaction

Select (“Mark something as interesting”)



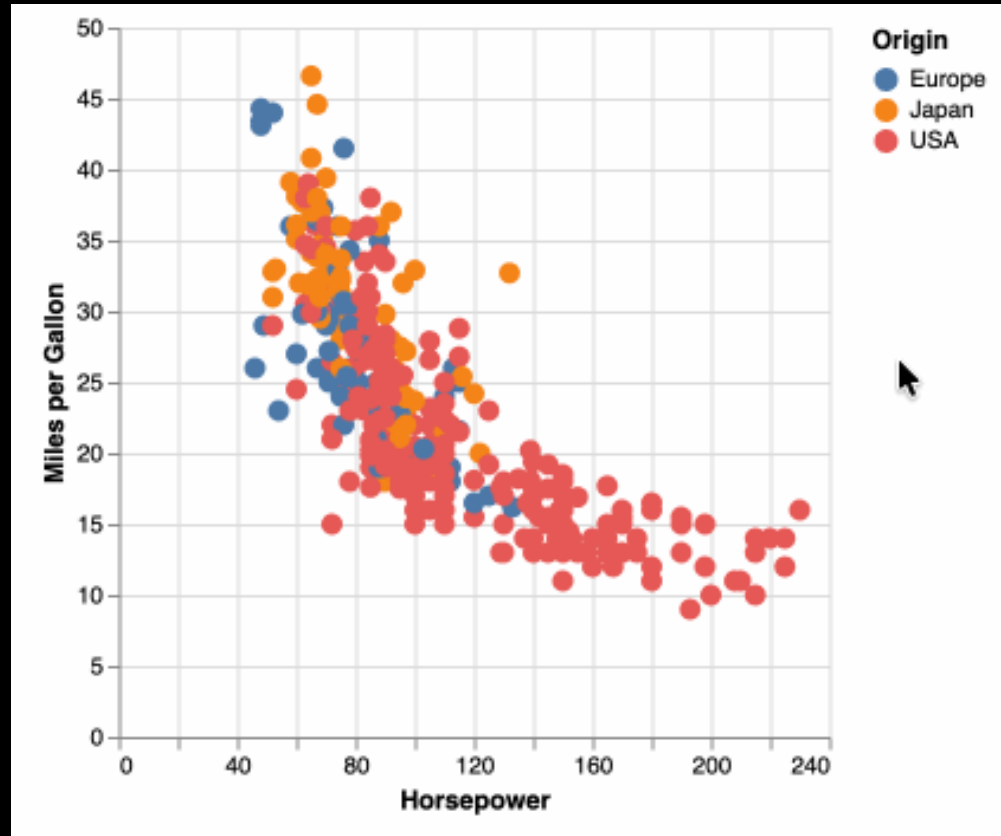
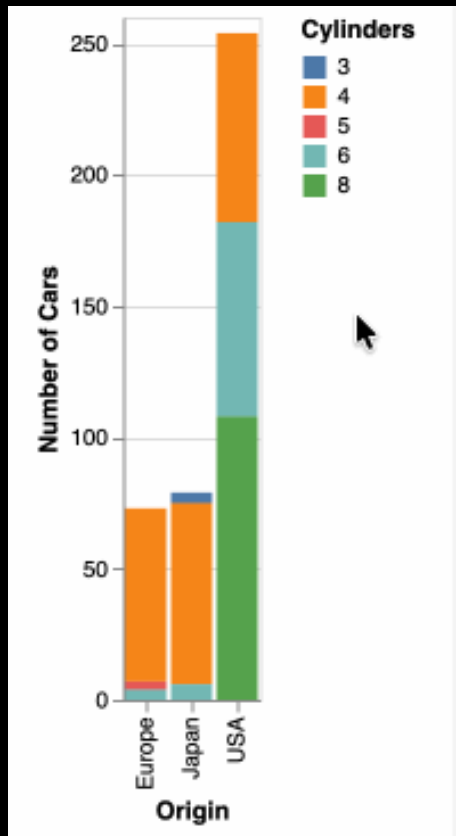
Select (“Mark something as interesting”)



Extensional selections:

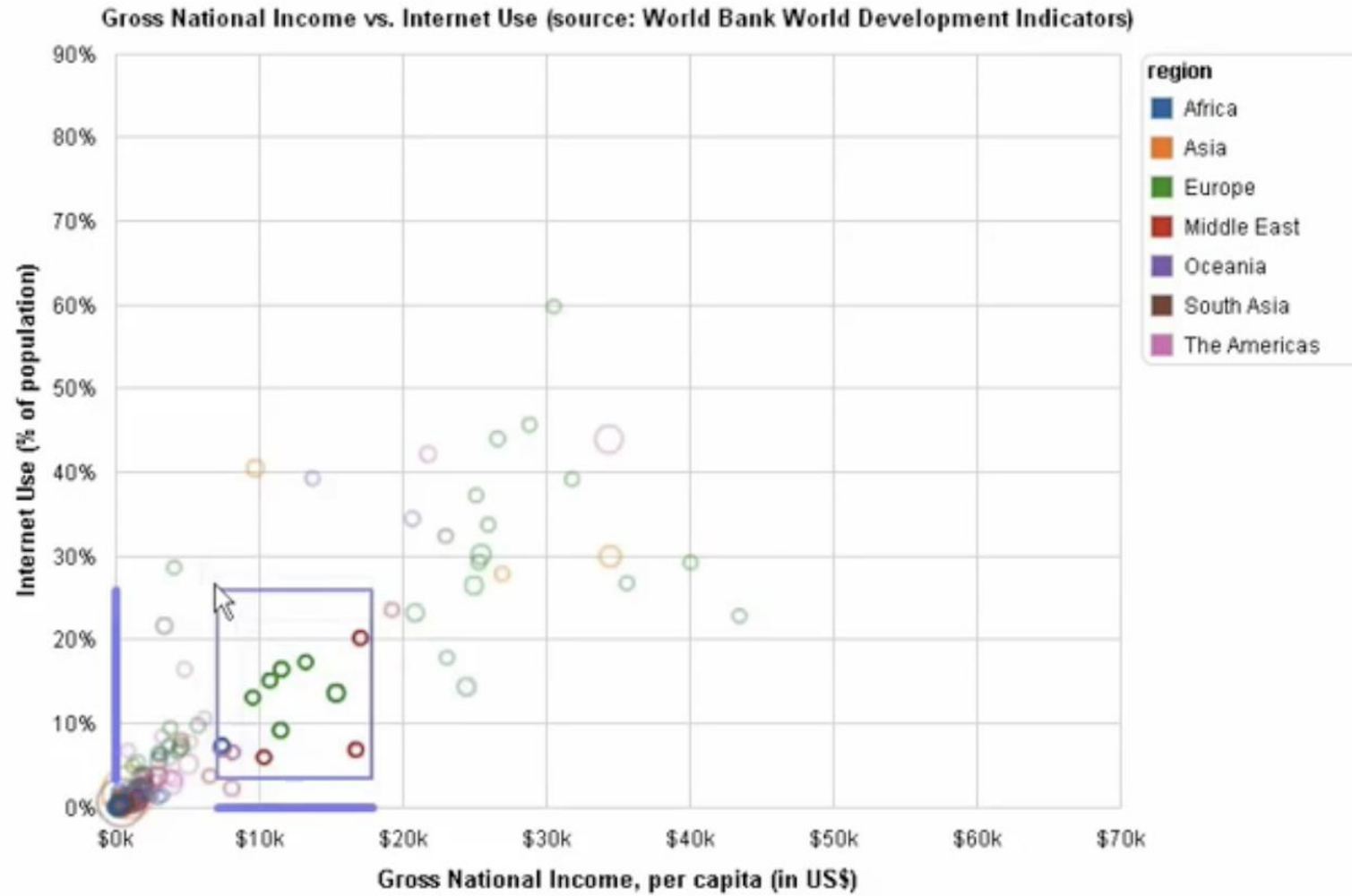
Items are selected by enumeration

Select (“Mark something as interesting”)



Intensional selections:

Items are selected by common properties

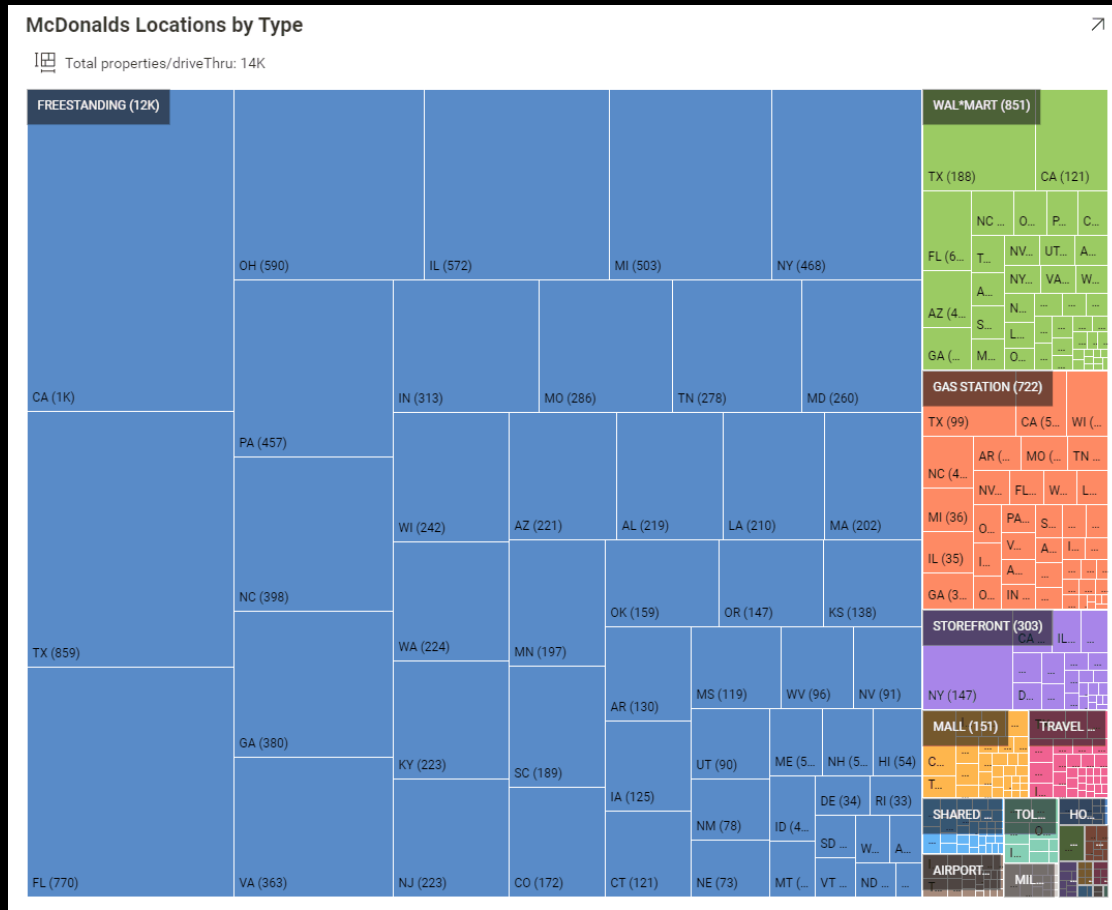


Generalized Selection via Interactive Query Relaxation
Heer et. al, 2008

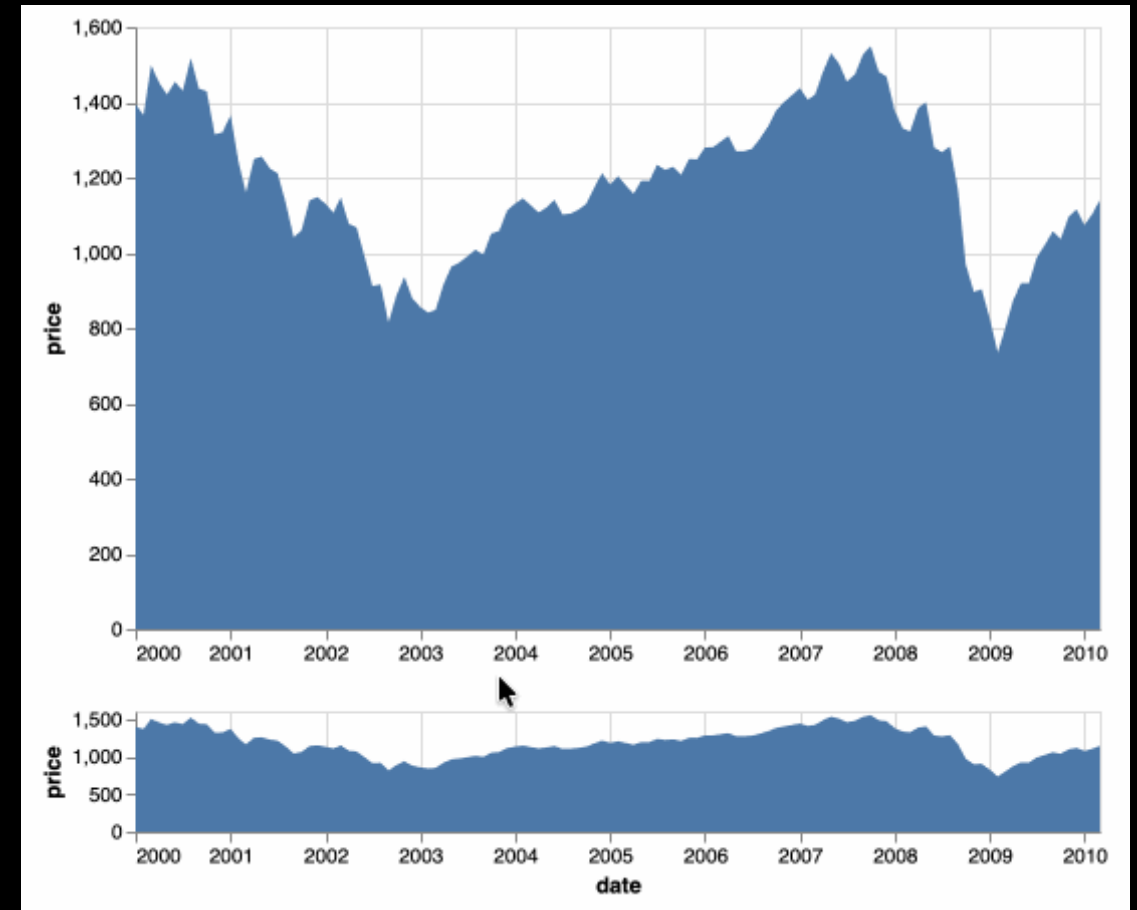
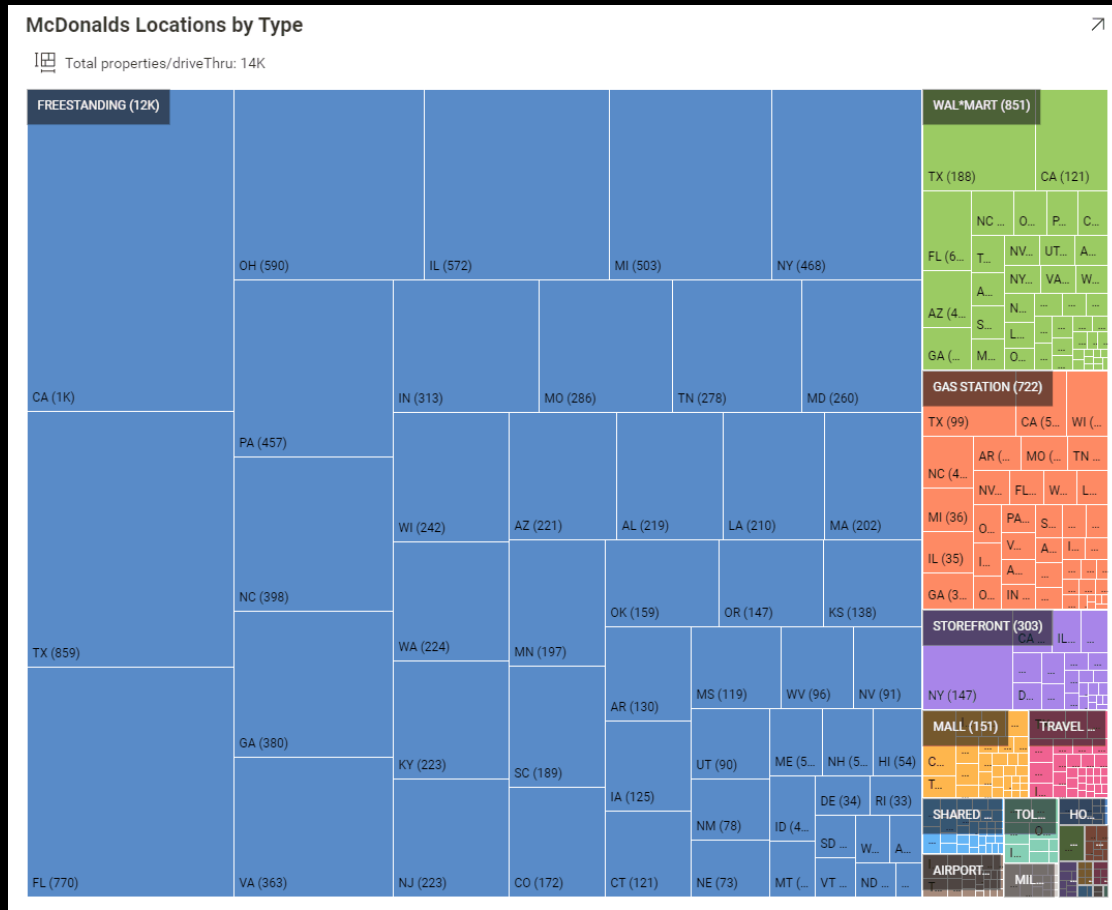
Abstract/elaborate (“Show me more or less detail”)

- Changing level of detail or abstraction

Abstract/elaborate (“Show me more or less detail”)



Abstract/elaborate (“Show me more or less detail”)

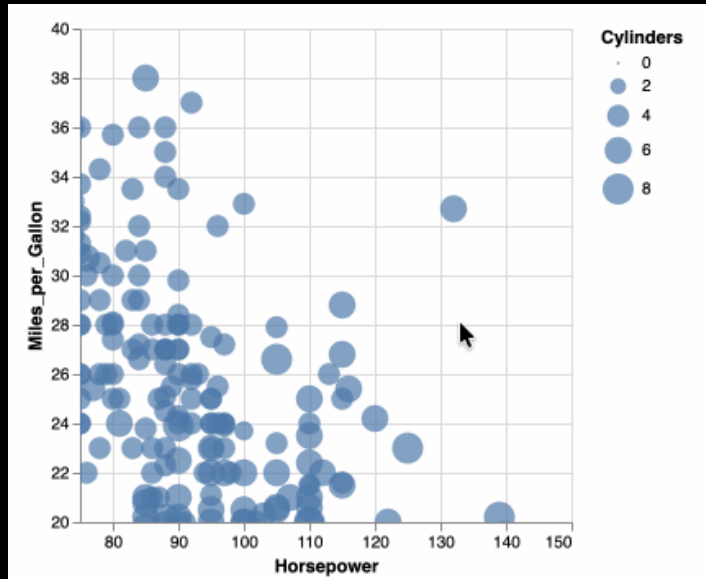


Explore (“Show me something else”)

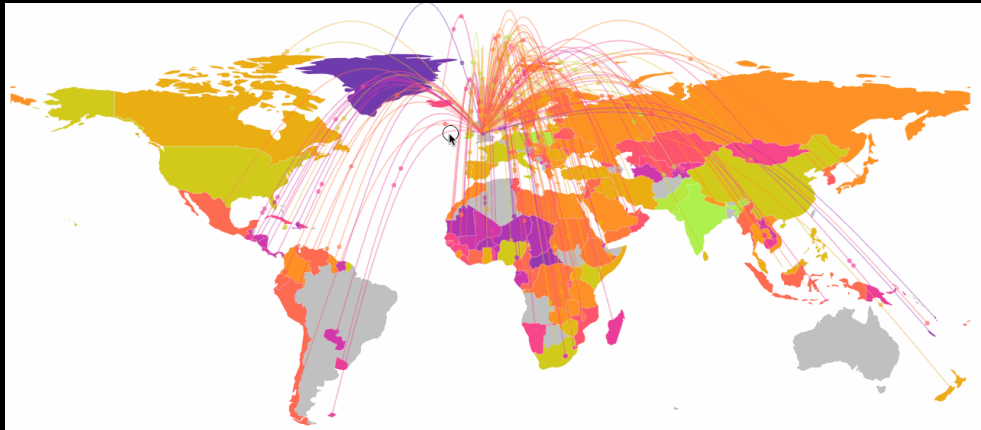
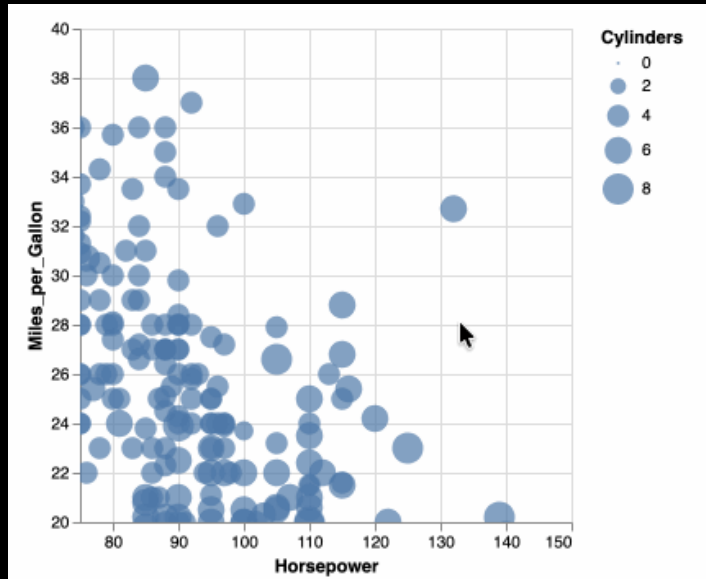
Enable users to examine a different subset of data

Overcome the limitation of display size

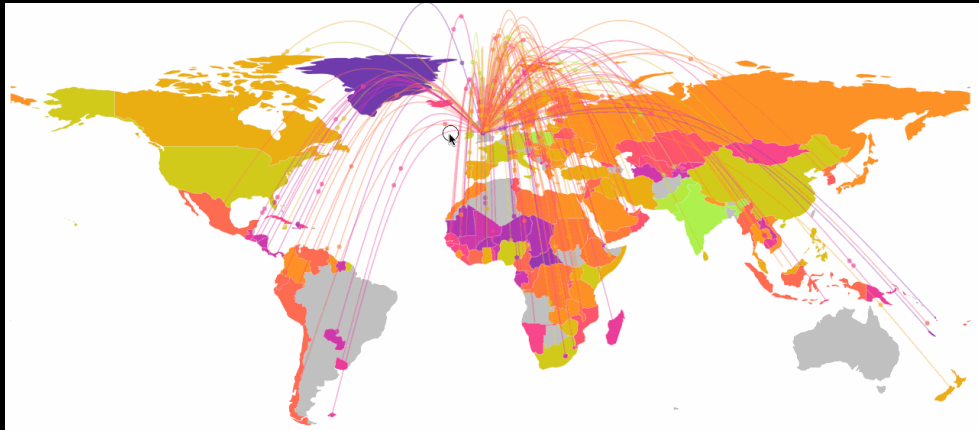
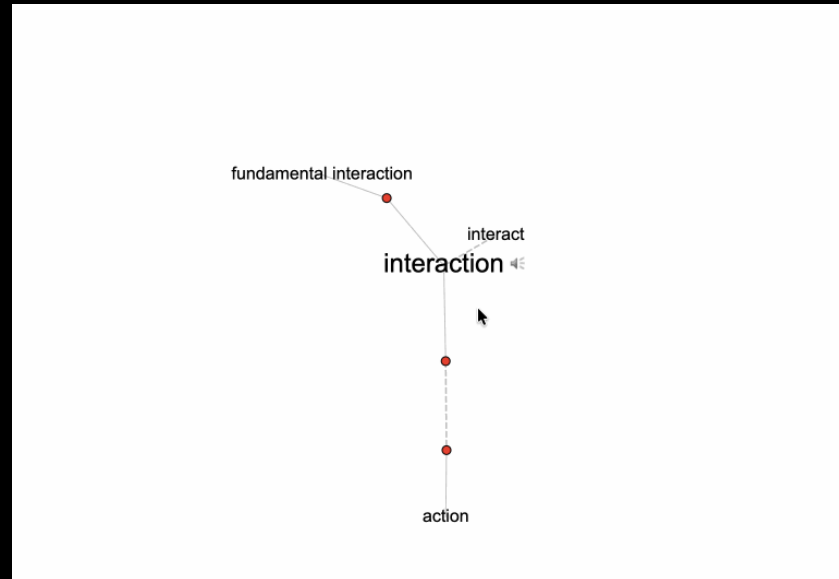
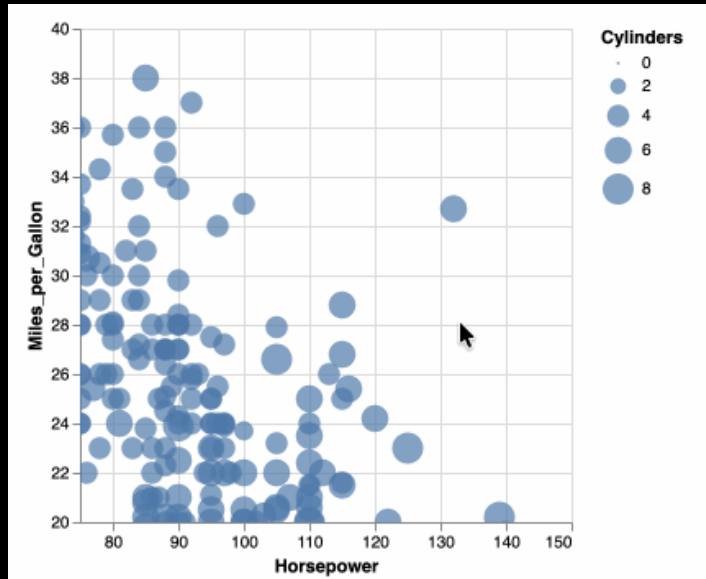
Explore (“Show me something else”)



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Explore (“Show me something else”)

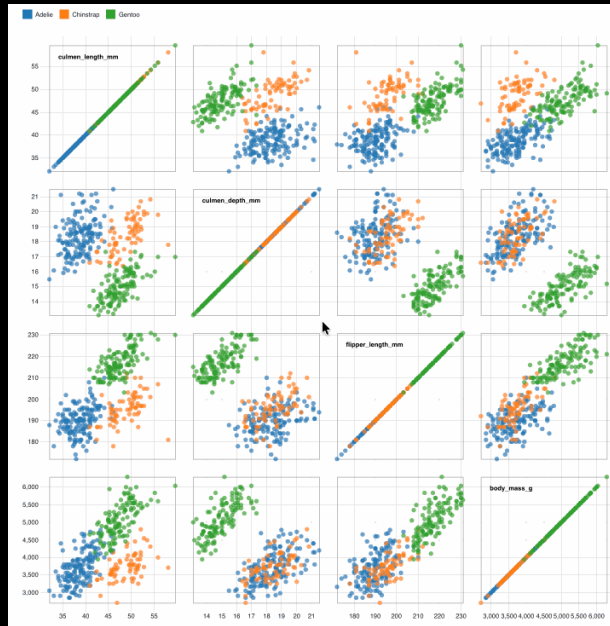


Connect (“Show me related items”)

Highlight associations and relationships

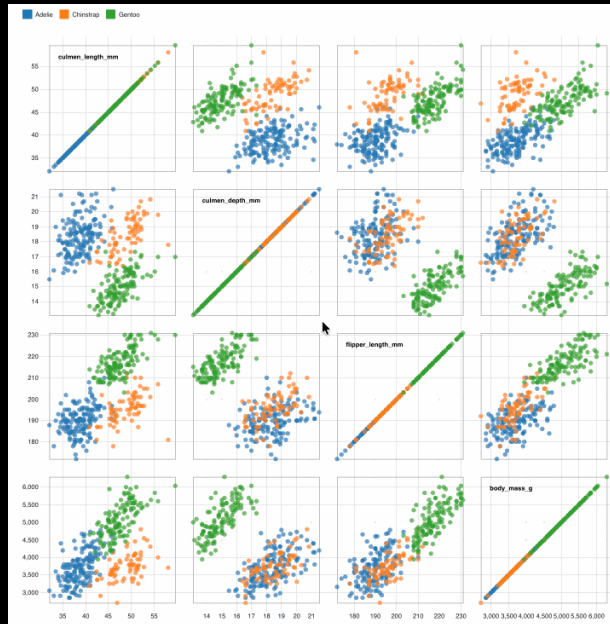
Show hidden data items that are relevant to a specified item

Connect (“Show me related items”)

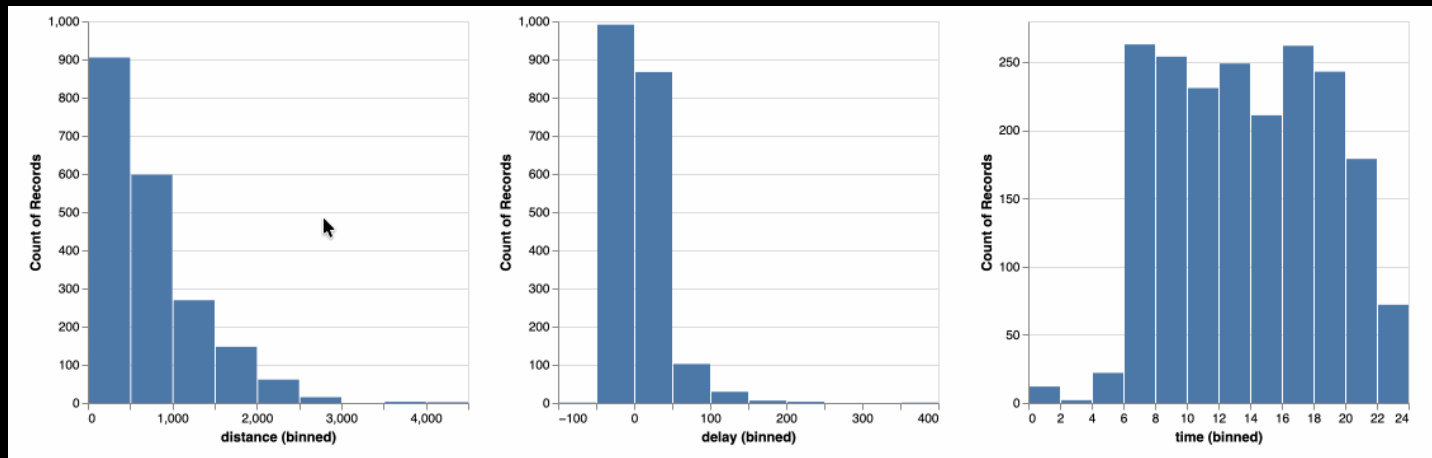


(link by tuple)

Connect (“Show me related items”)

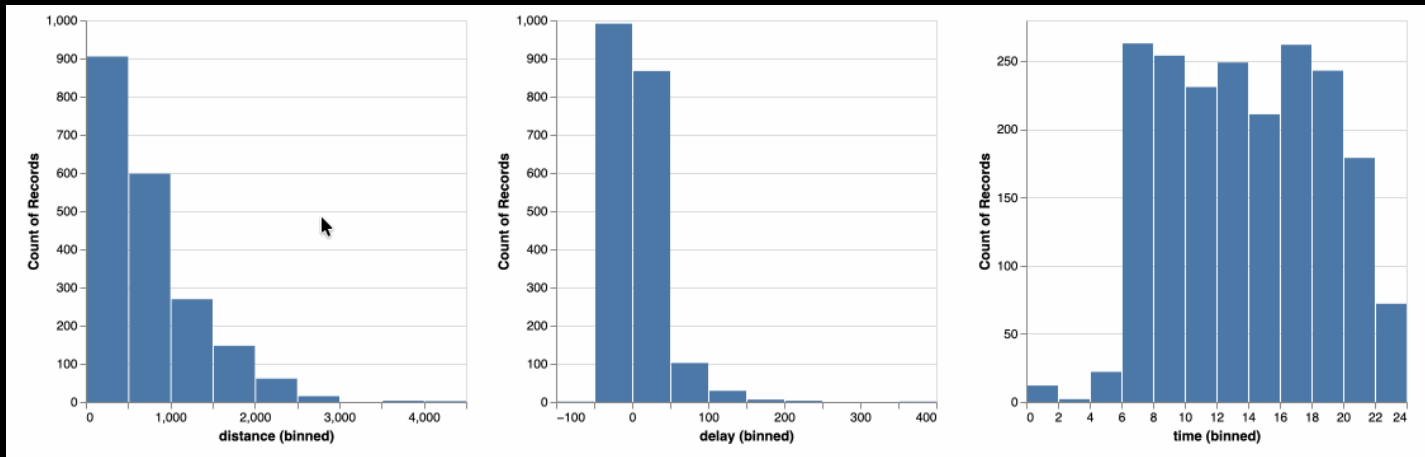
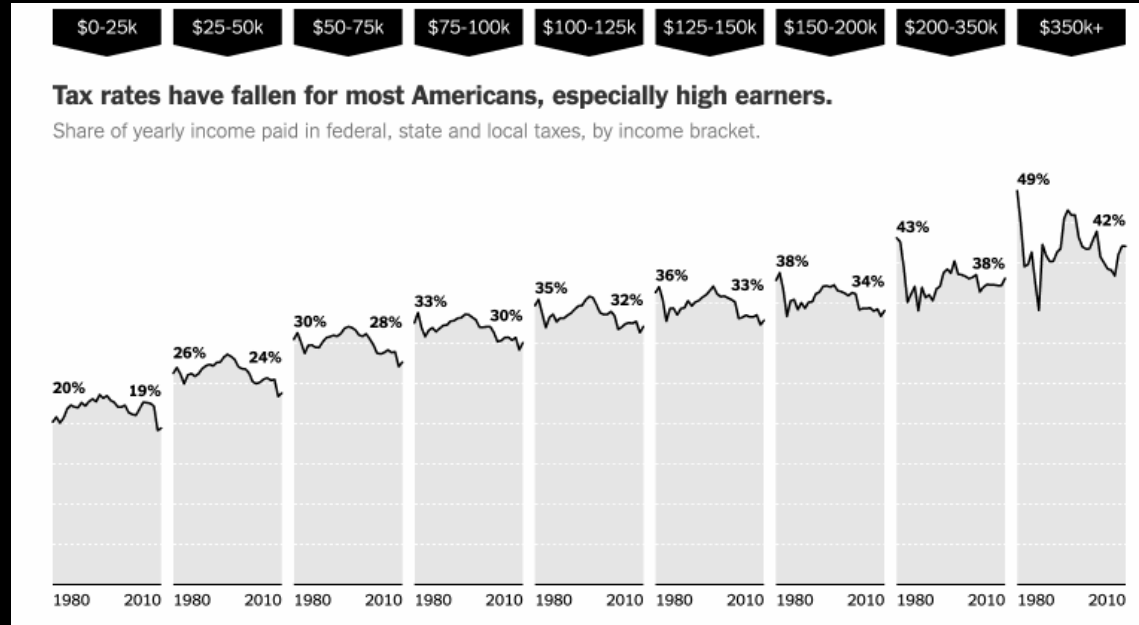
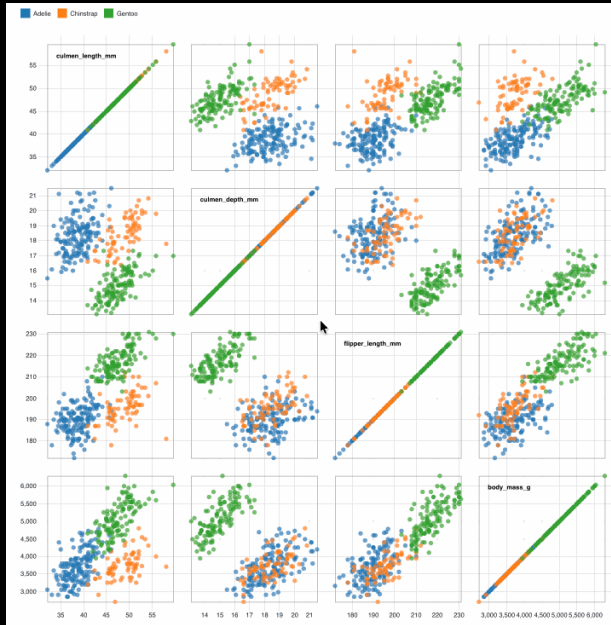


(link by tuple)

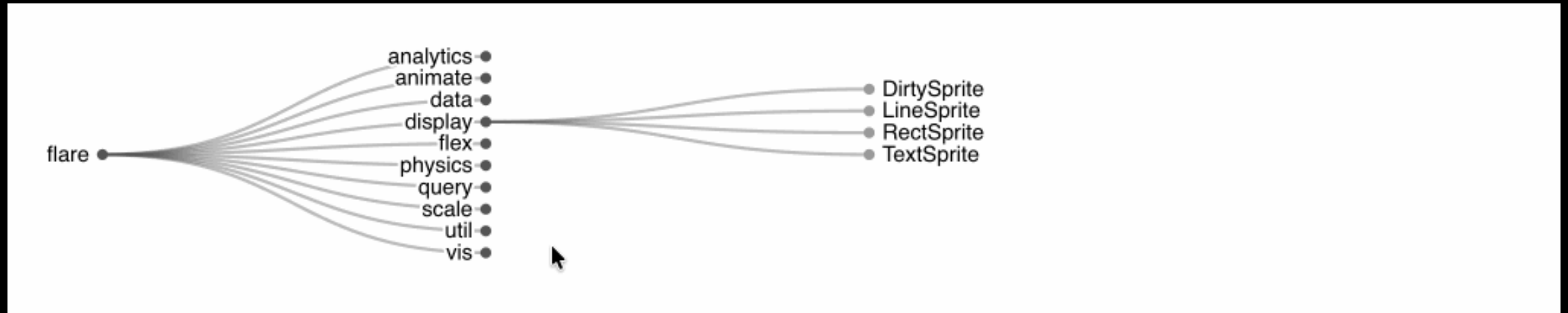


(link by query)

Connect (“Show me related items”)



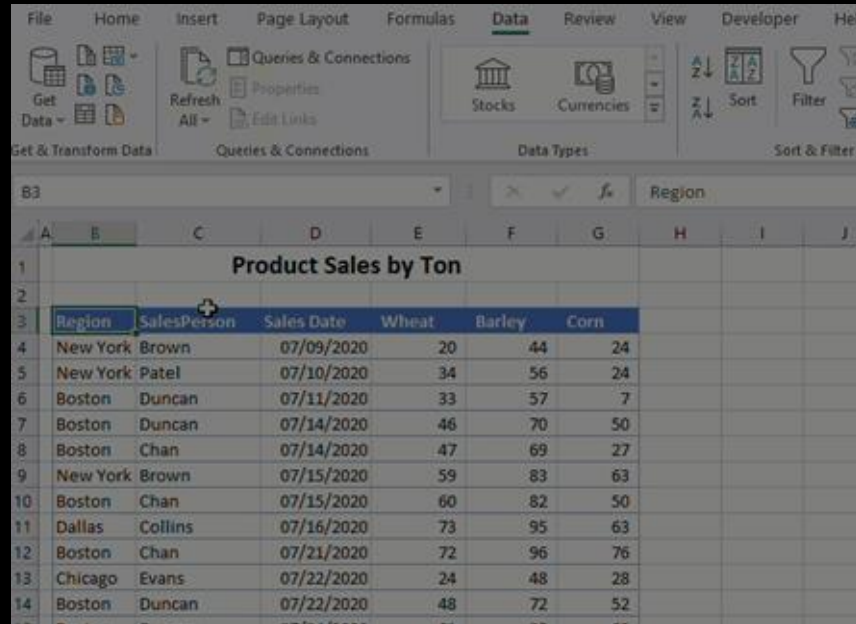
Connect (“Show me related items”)



Reconfigure (“Show me a different arrangement”)

Provide different perspectives by changing the spatial arrangement of representation

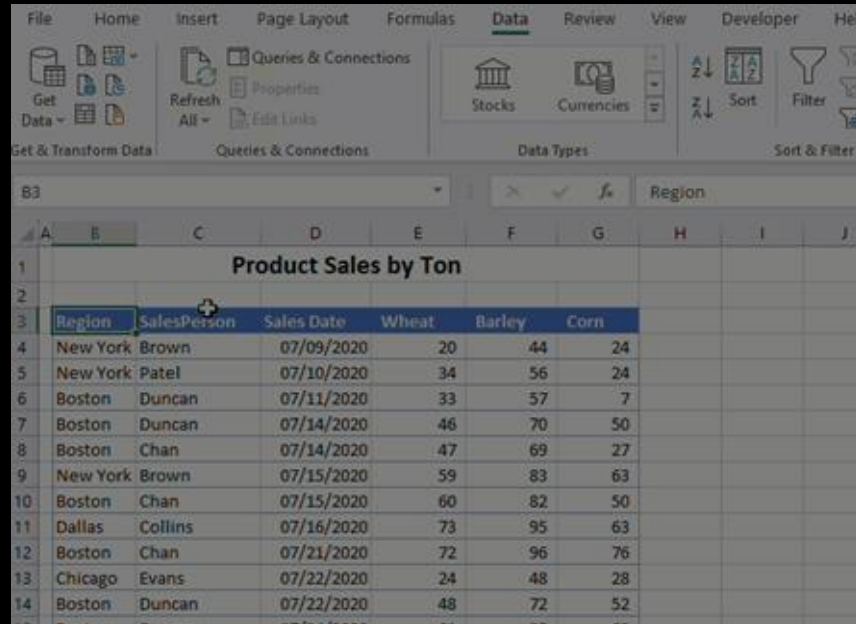
Reconfigure (“Show me a different arrangement”)



The screenshot shows the Microsoft Excel interface with the 'Data' tab selected in the ribbon. The ribbon includes sections for 'Get & Transform Data', 'Queries & Connections', 'Data Types', and 'Sort & Filter'. The 'Sort & Filter' section is expanded, showing 'Sort' and 'Filter' options. The active worksheet contains a table titled 'Product Sales by Ton' with the following data:

Region	SalesPerson	Sales Date	Wheat	Barley	Corn
New York	Brown	07/09/2020	20	44	24
New York	Patel	07/10/2020	34	56	24
Boston	Duncan	07/11/2020	33	57	7
Boston	Duncan	07/14/2020	46	70	50
Boston	Chan	07/14/2020	47	69	27
New York	Brown	07/15/2020	59	83	63
Boston	Chan	07/15/2020	60	82	50
Dallas	Collins	07/16/2020	73	95	63
Boston	Chan	07/21/2020	72	96	76
Chicago	Evans	07/22/2020	24	48	28
Boston	Duncan	07/22/2020	48	72	52

Reconfigure (“Show me a different arrangement”)



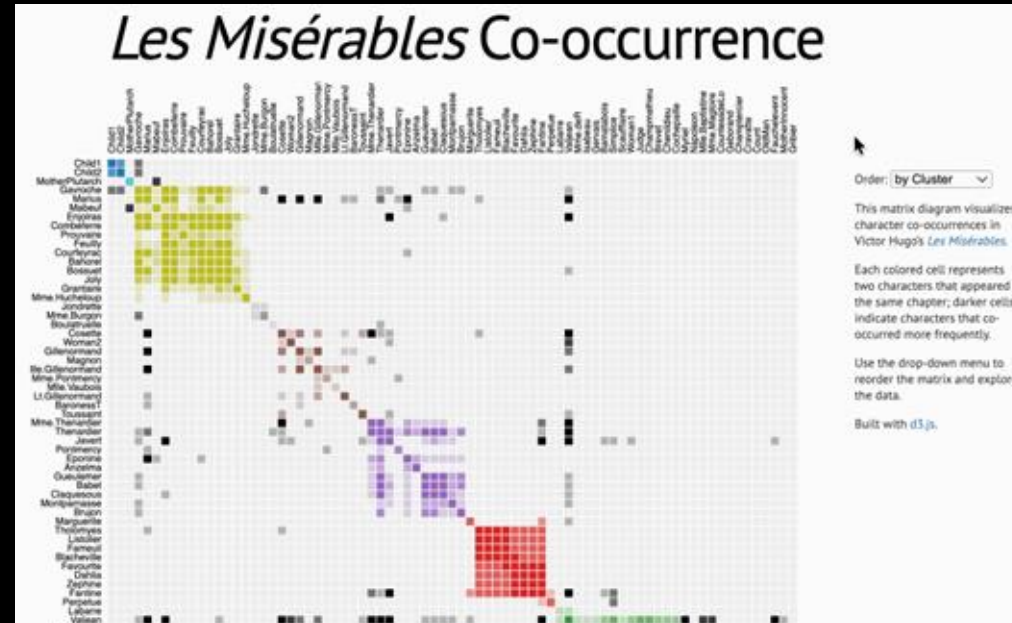
Product Sales by Ton						
Region	SalesPerson	Sales Date	Wheat	Barley	Corn	
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New York	Patel	07/10/2020	34	56	24	
Boston	Duncan	07/11/2020	33	57	7	
Boston	Duncan	07/14/2020	46	70	50	
Boston	Chan	07/14/2020	47	69	27	
New York	Brown	07/15/2020	59	83	63	
Boston	Chan	07/15/2020	60	82	50	
Dallas	Collins	07/16/2020	73	95	63	
Boston	Chan	07/21/2020	72	96	76	
Chicago	Evans	07/22/2020	24	48	28	
Boston	Duncan	07/22/2020	48	72	52	

PET NAME	PARENT NAME	EMAIL	LAST VISIT
swift	joy ward	blanca_bartoletti7@hotmail.com	1/4/2019
raynor	micaela volkman	thora.ohara@gmail.com	10/2/2019
predovic	caesar towne	clare_rogahn@hotmail.com	1/15/2019
yost	kennith ullrich	dallas.kassulke47@hotmail.com	1/5/2019
feil	regan macgyver	fanny67@gmail.com	7/2/2019

Reconfigure (“Show me a different arrangement”)

Product Sales by Ton						
Region	Product	Ton	Price	Weight	Volume	Value
New York	Brown	07/08/2020	20	44	24	
New York	Patel	07/10/2020	34	56	24	
Boston	Duncan	07/11/2020	83	57	7	
Boston	Duncan	07/14/2020	46	70	50	
Boston	Chan	07/14/2020	47	69	27	
New York	Brown	07/15/2020	55	83	63	
Boston	Chan	07/15/2020	60	82	50	
Dallas	Collins	07/16/2020	73	95	63	
Boston	Chan	07/21/2020	72	96	76	
Chicago	Evans	07/22/2020	24	48	28	
Boston	Duncan	07/24/2020	48	72	52	

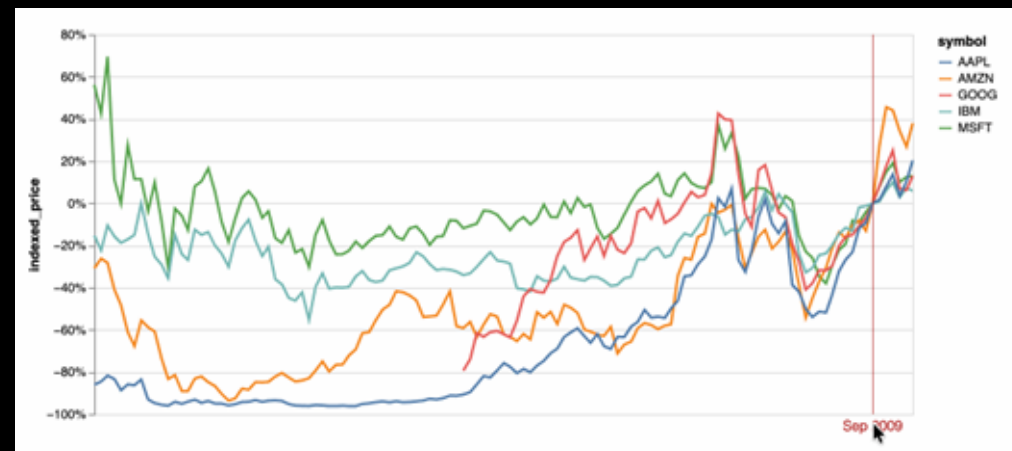
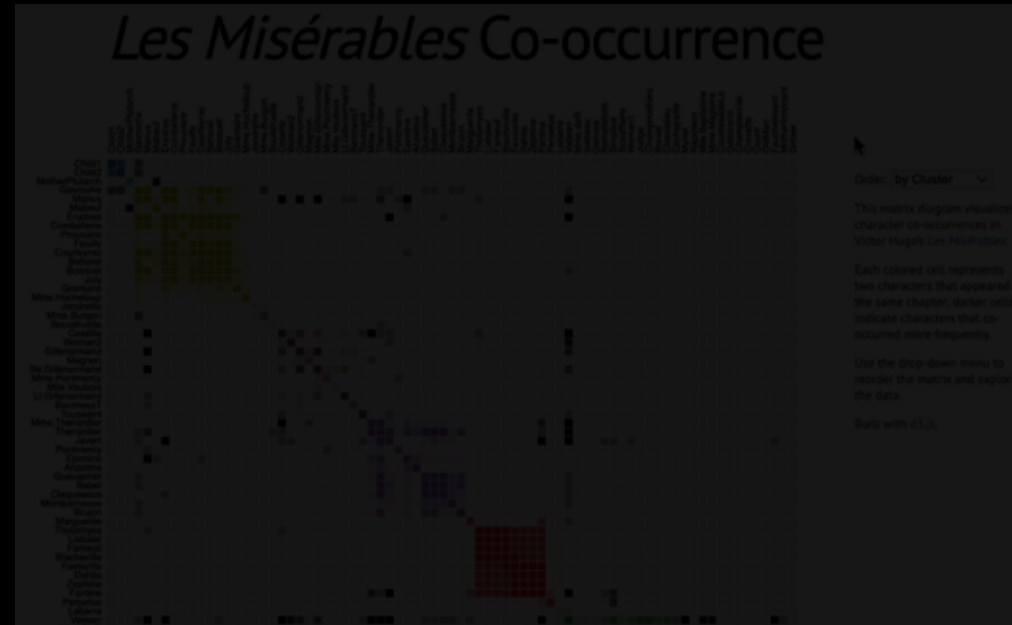
PET NAME	PARENT NAME	EMAIL	LAST VISIT
swift	joy ward	blanca_bartoletti7@hotmail.com	1/4/2019
raynor	micaela volkman	thora.ohara@gmail.com	10/2/2019
predovic	caesar towne	clare_roahn@hotmail.com	1/15/2019
yost	kennith ulrich	dallas.kassulke47@hotmail.com	1/5/2019
feil	regan macgyver	fanny67@gmail.com	7/2/2019



Reconfigure (“Show me a different arrangement”)

Product Sales by Ton					
Region	Product	Date	Ton	Price	Value
New York	Brown	07/08/2020	20	44	24
New York	Patel	07/10/2020	34	56	24
Boston	Duncan	07/11/2020	83	57	7
Boston	Duncan	07/14/2020	46	70	50
Boston	Chan	07/14/2020	47	69	27
New York	Brown	07/15/2020	33	83	63
Boston	Chan	07/15/2020	60	82	50
Dallas	Collins	07/16/2020	73	95	62
Boston	Chan	07/21/2020	72	96	76
Chicago	Evans	07/22/2020	24	48	28
Boston	Duncan	07/24/2020	48	72	52

PET NAME	PARENT NAME	EMAIL	LAST VISIT
swift	joy ward	blanca_bartoletti7@hotmail.com	1/4/2019
raynor	micaela volkman	thora.ohara@gmail.com	10/2/2019
predovic	caesar towne	clare_roahn@hotmail.com	1/15/2019
yost	kennith ulrich	dallas.kassulke47@hotmail.com	1/5/2019
feil	regan macgyver	fanny67@gmail.com	7/2/2019

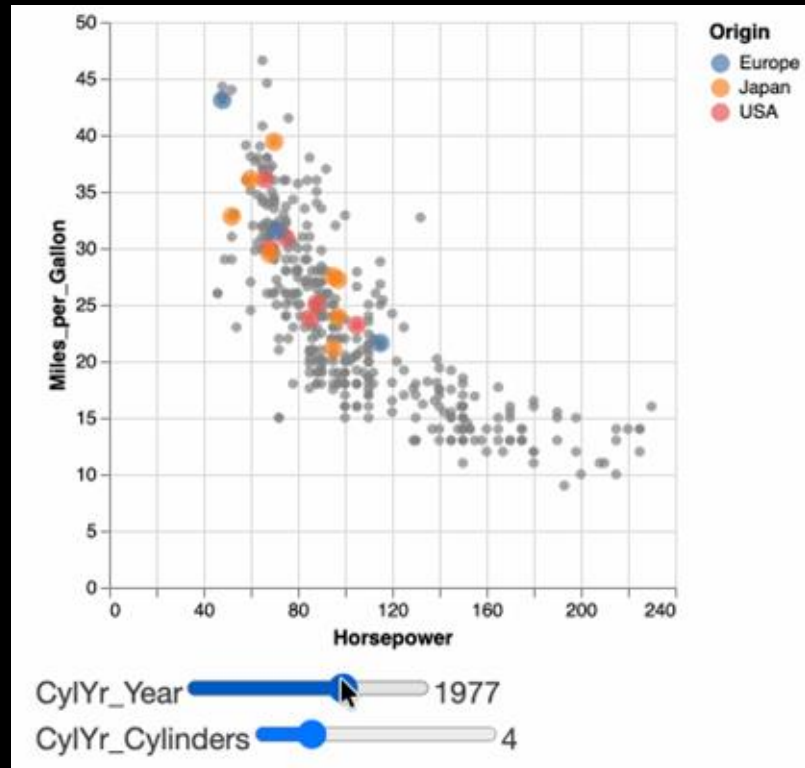


Filter (“Show me something conditionally”)

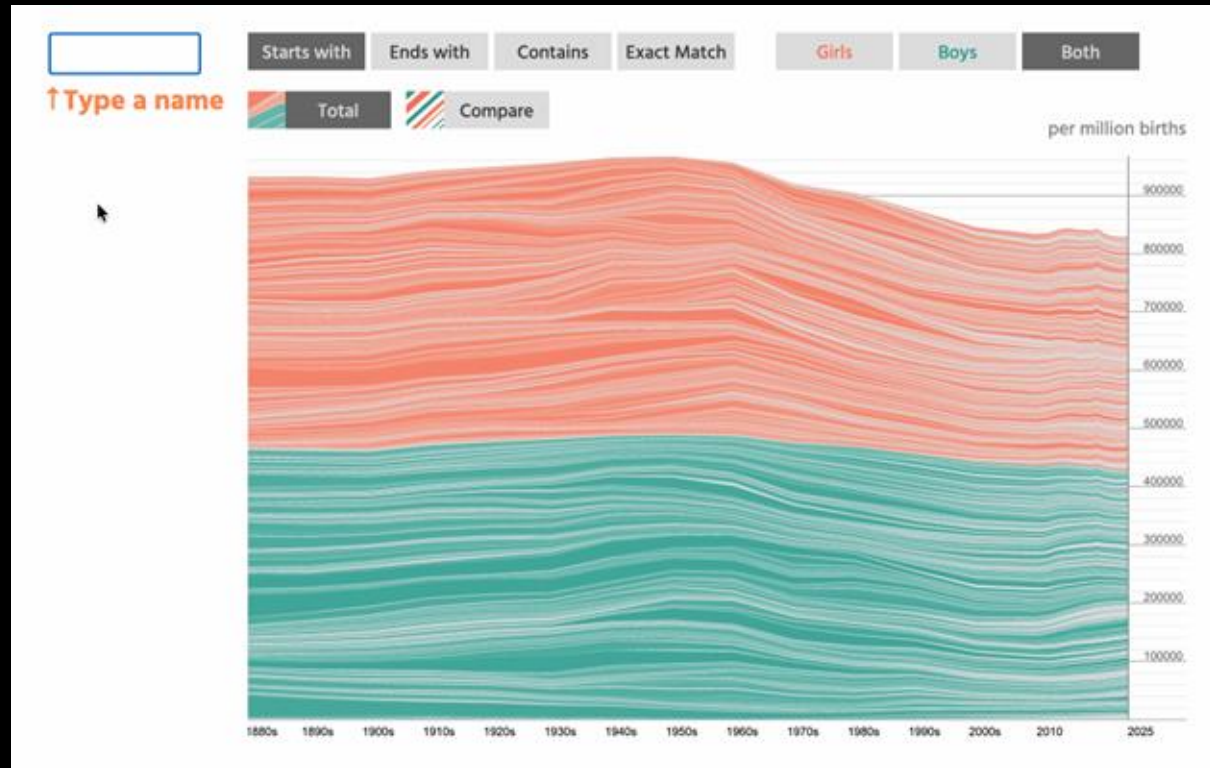
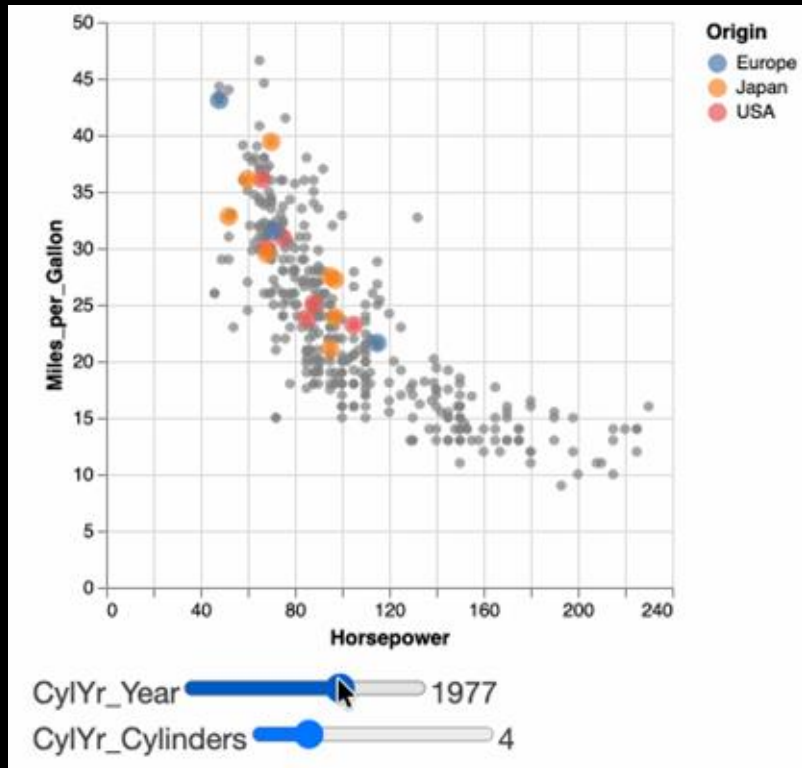
Changing the set of data cases being displayed. Useful for targeted analysis or even limiting data for performance (e.g., rendering)

Probably the most common interaction you will run into when designing data systems.

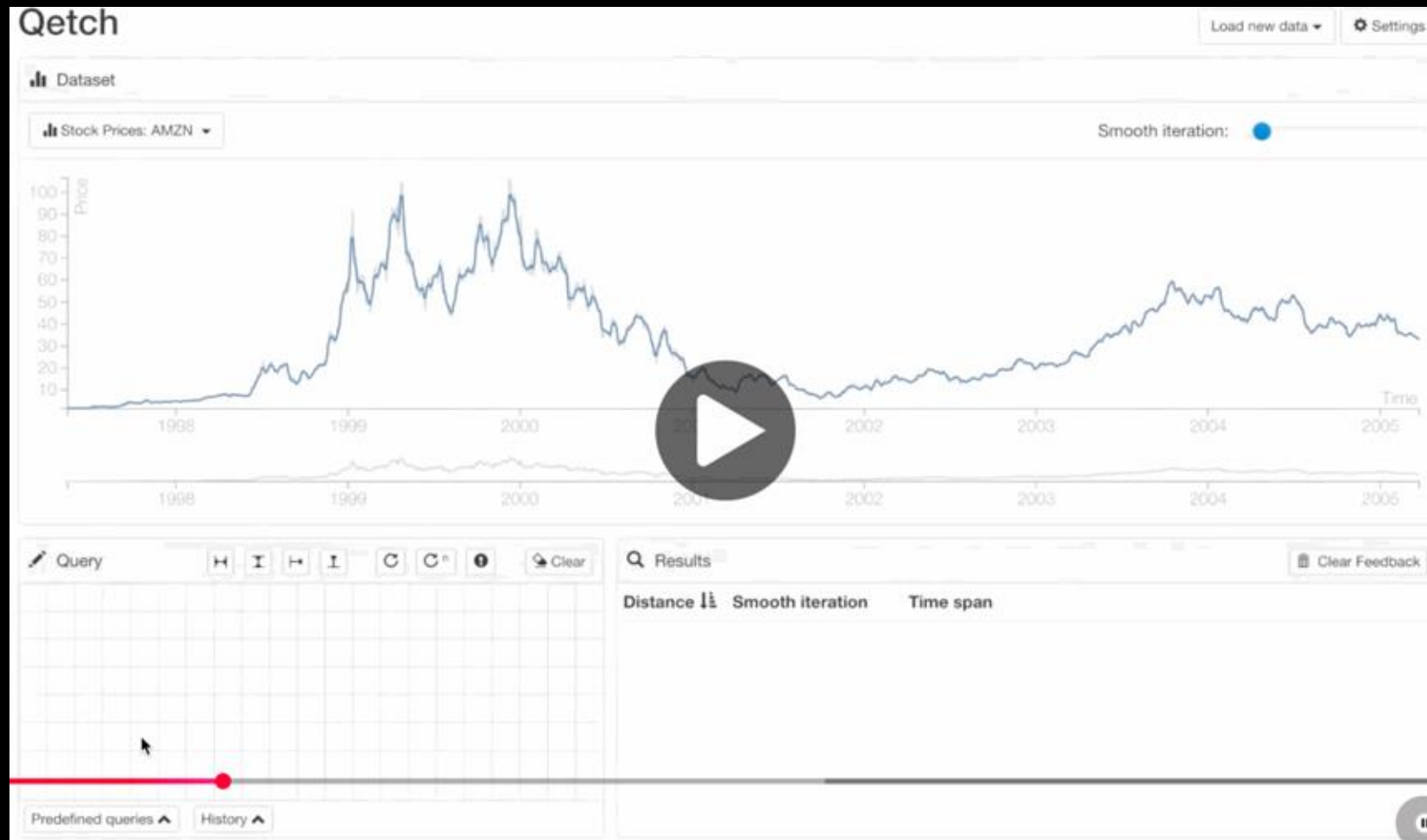
Filter (“Show me something conditionally”)



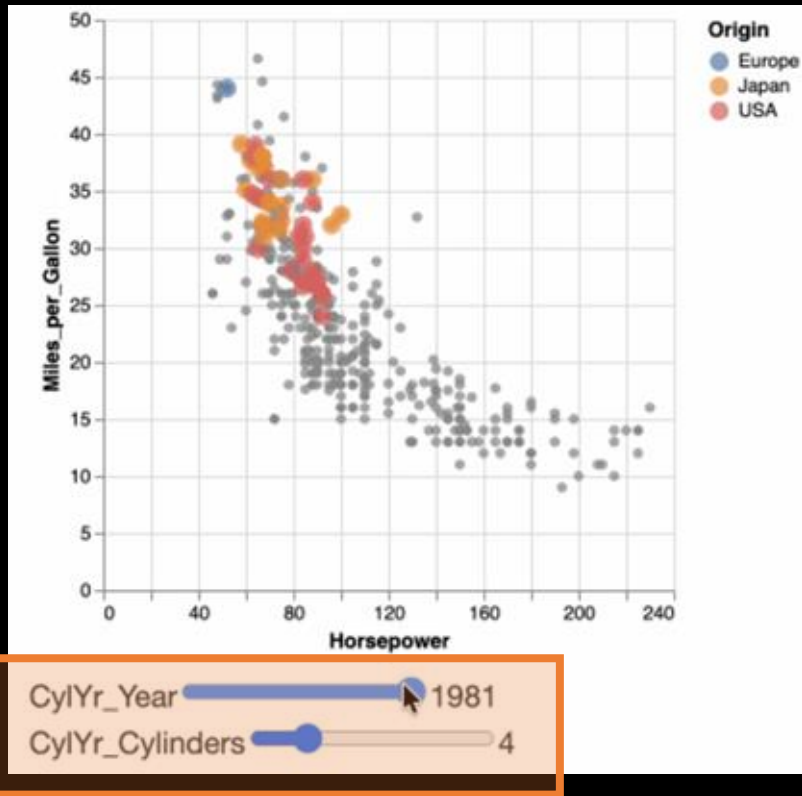
Filter (“Show me something conditionally”)



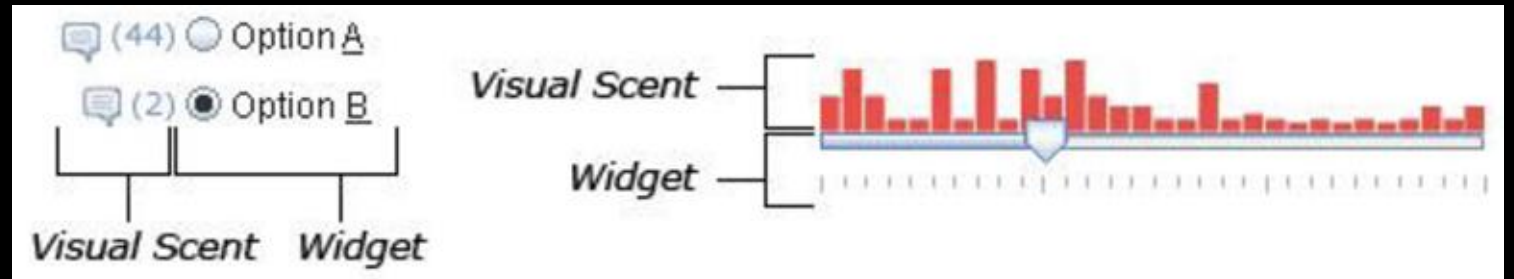
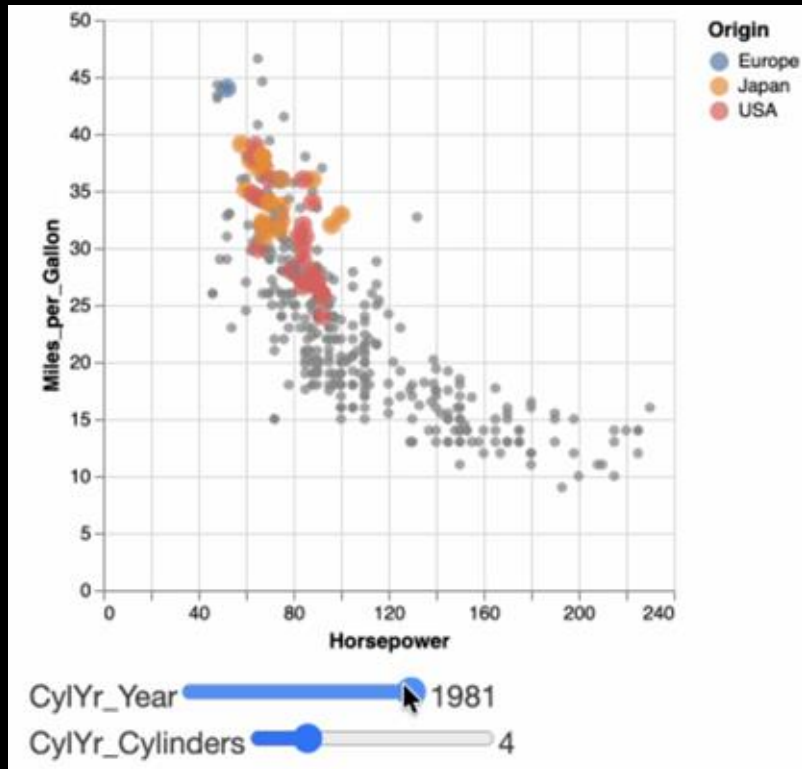
Filter (“Show me something conditionally”)



Filter (“Show me something conditionally”)

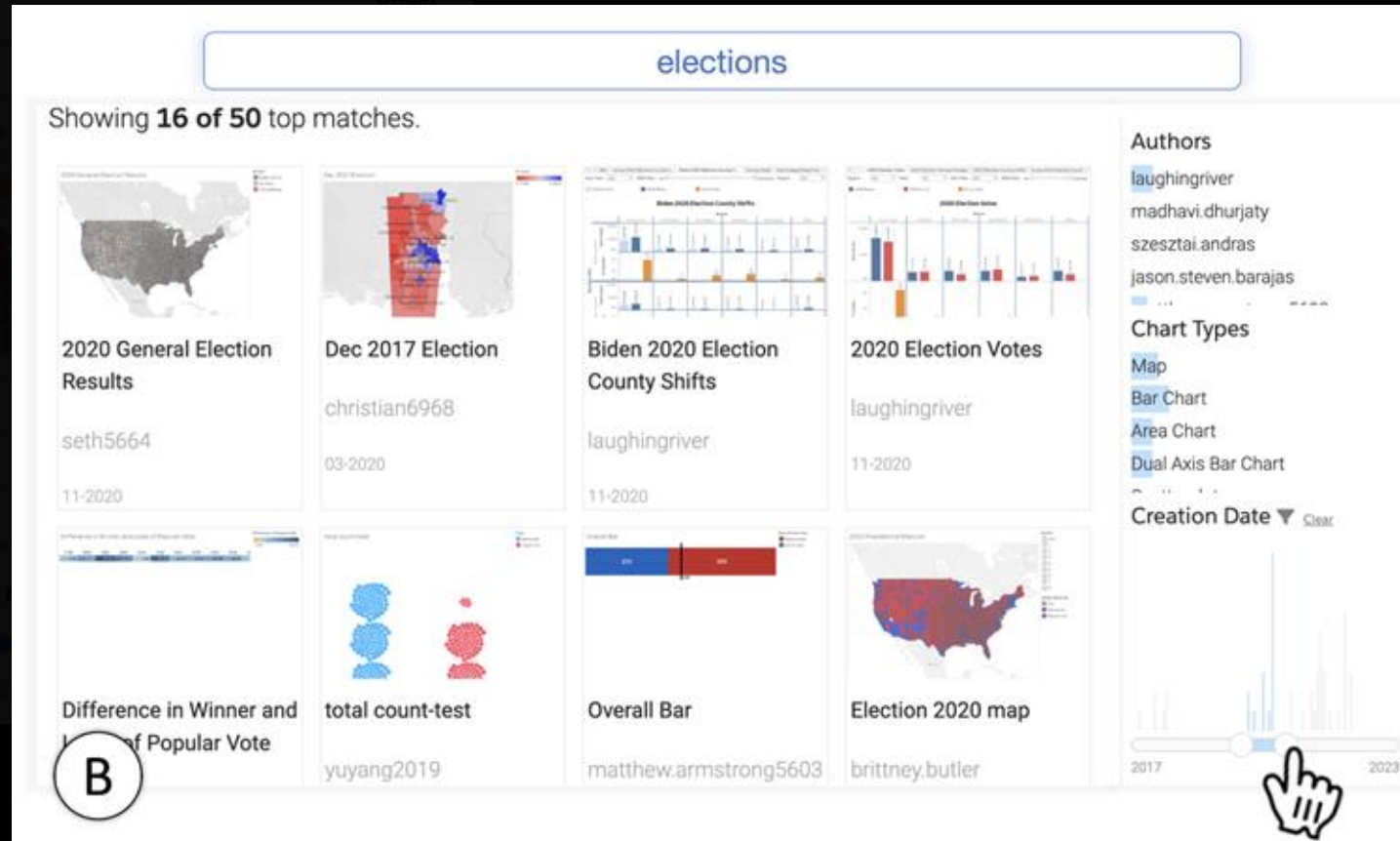


Filter (“Show me something conditionally”)



Scented Widgets: Improving Navigation Cues with Embedded Visualizations
Willet et. al, 2007

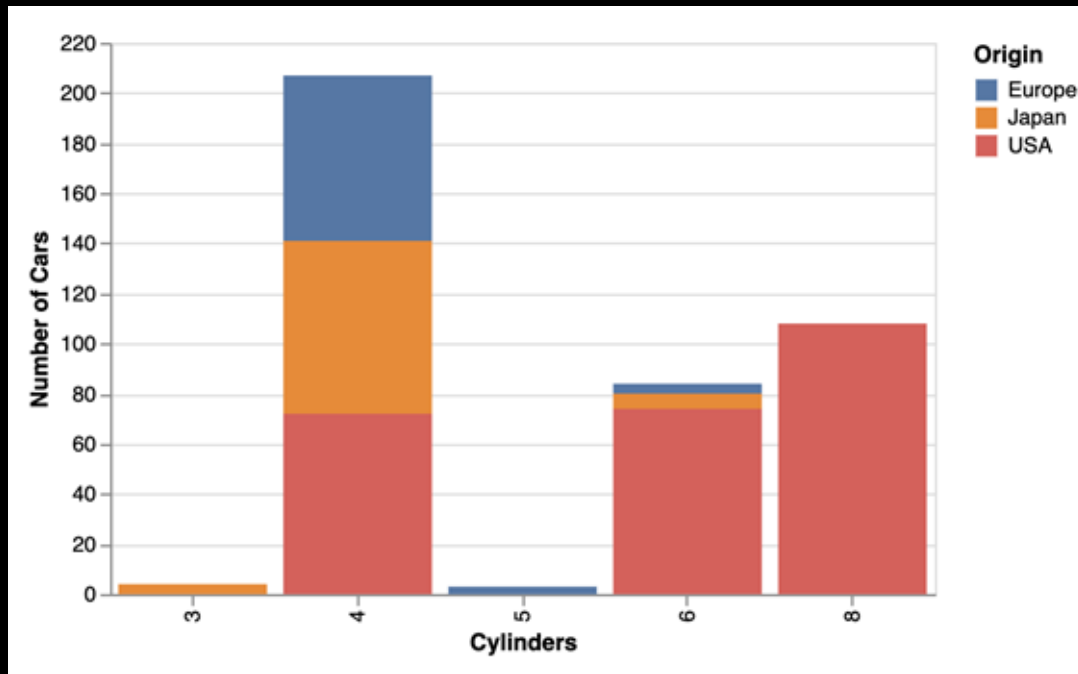
Filter (“Show me something conditionally”)



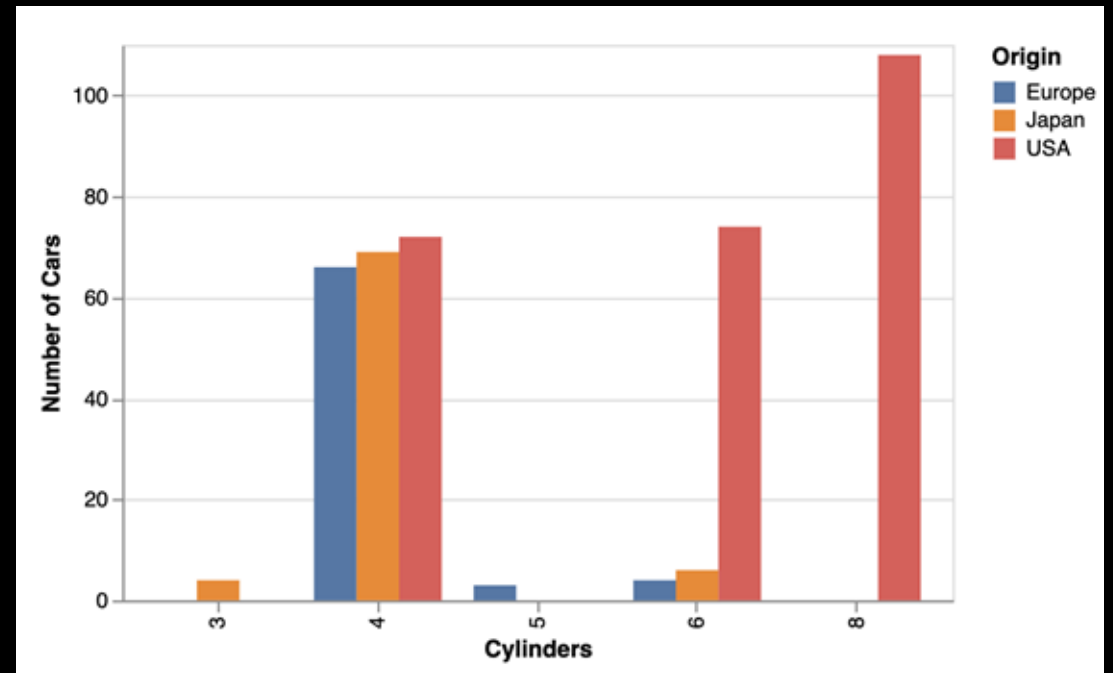
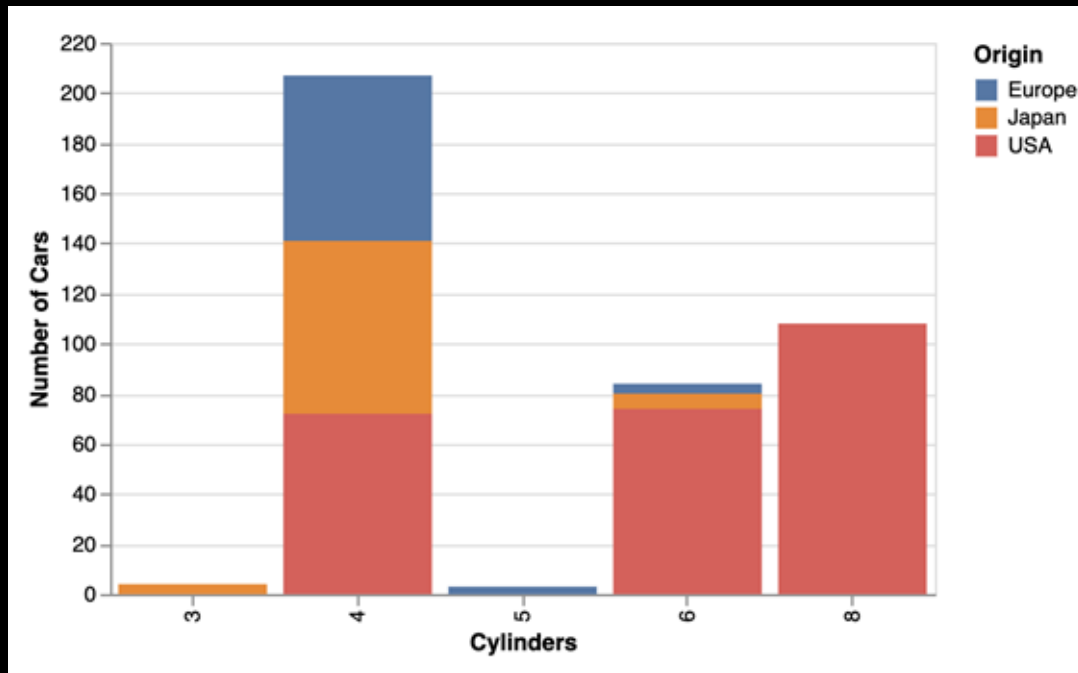
Encode (“Show me a different visual representation”)

Use visual mappings to saliently identify/highlight specific types of patterns

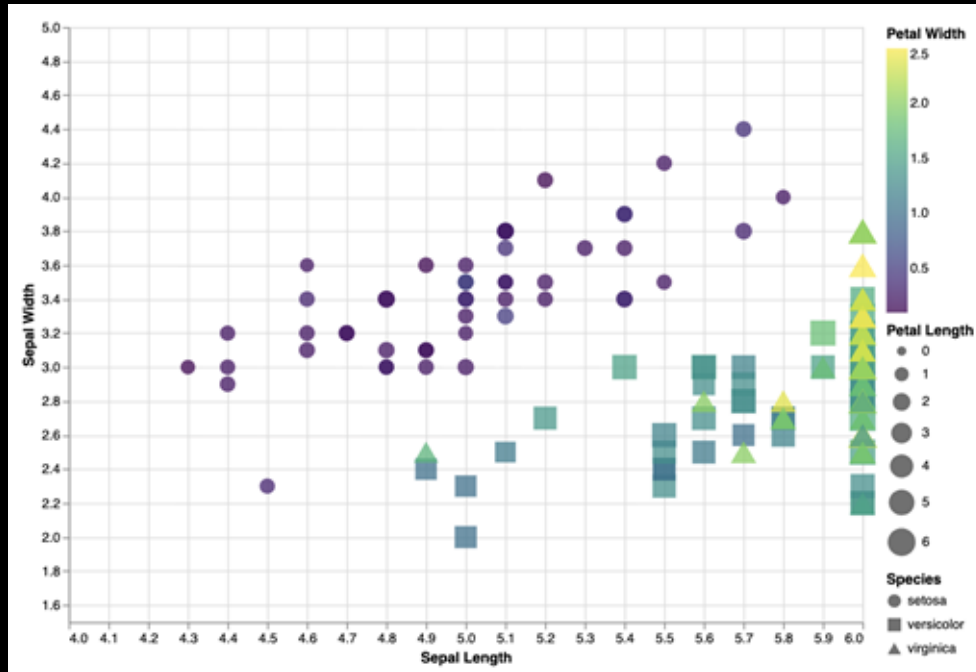
Encode (“Show me a different visual representation”)



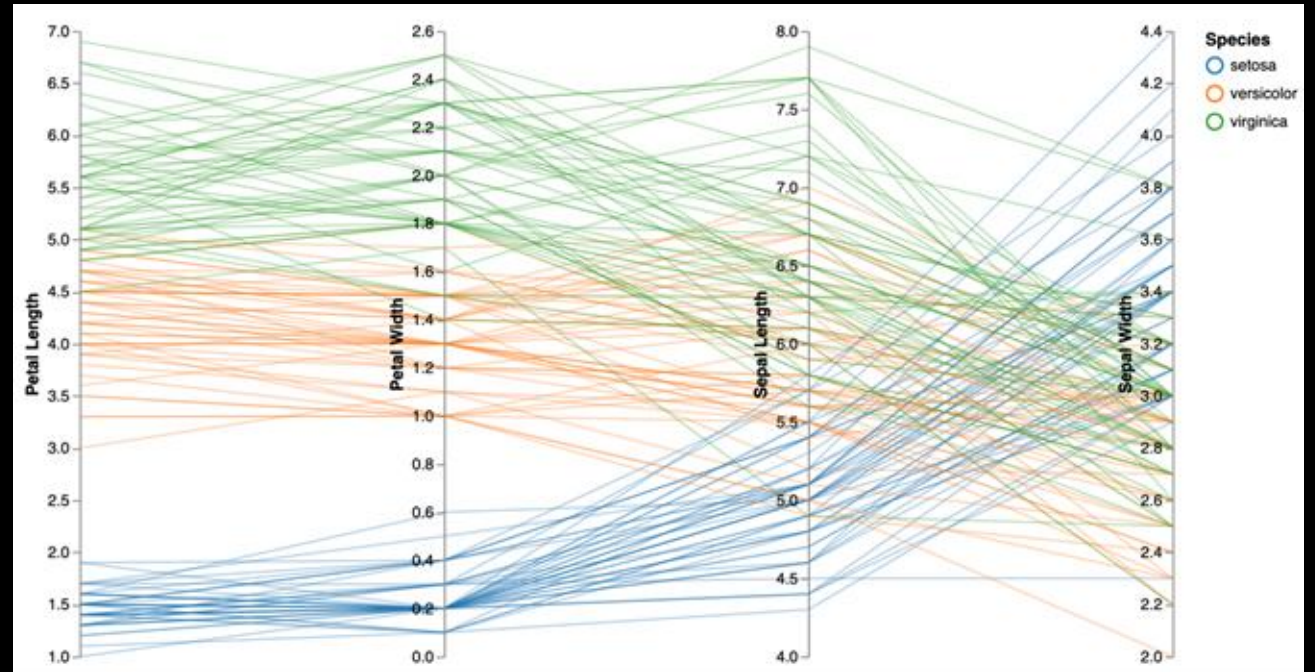
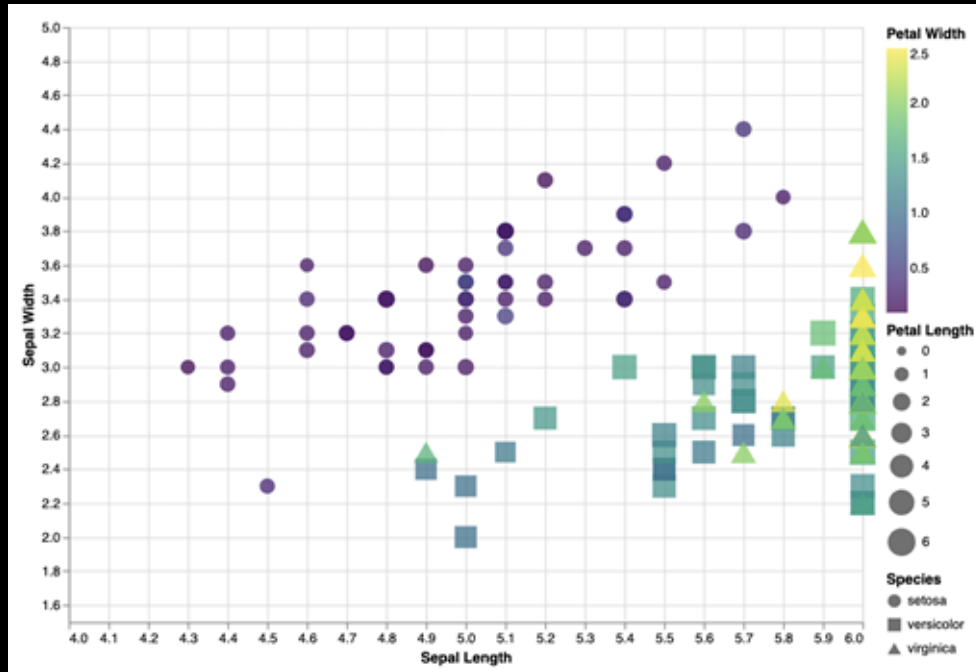
Encode (“Show me a different visual representation”)



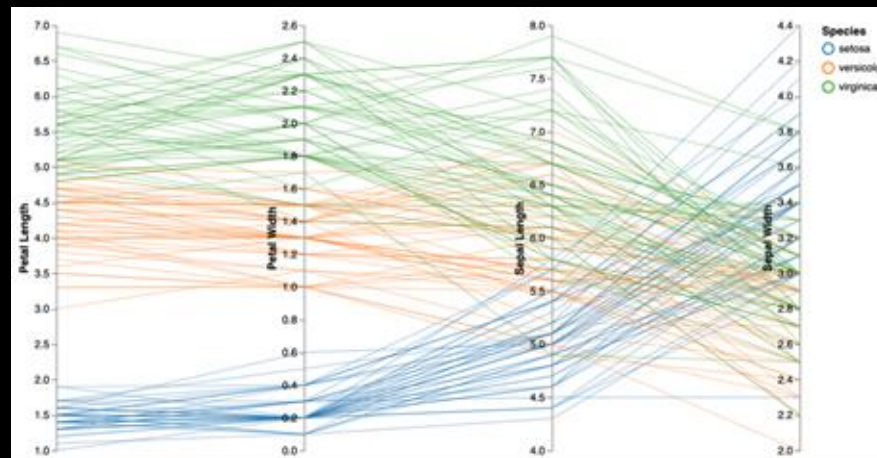
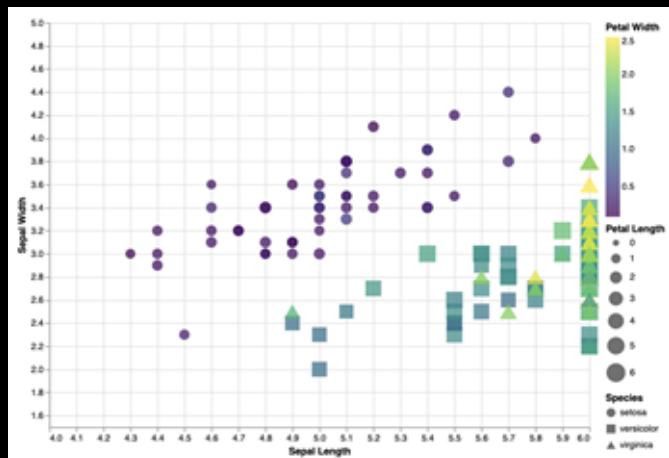
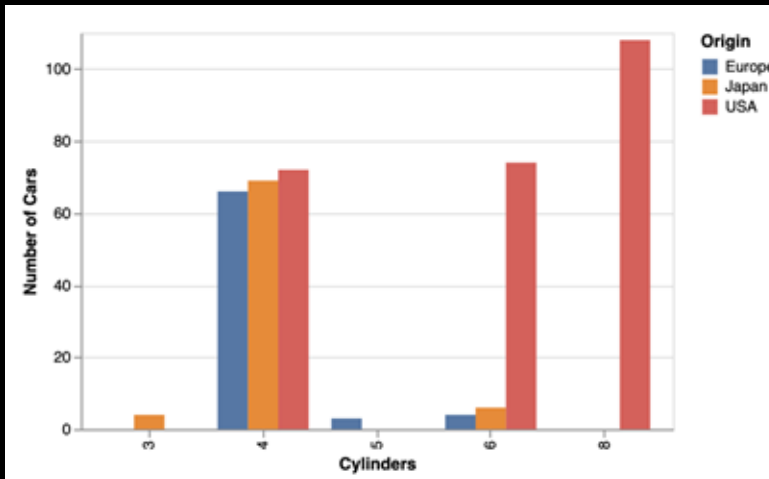
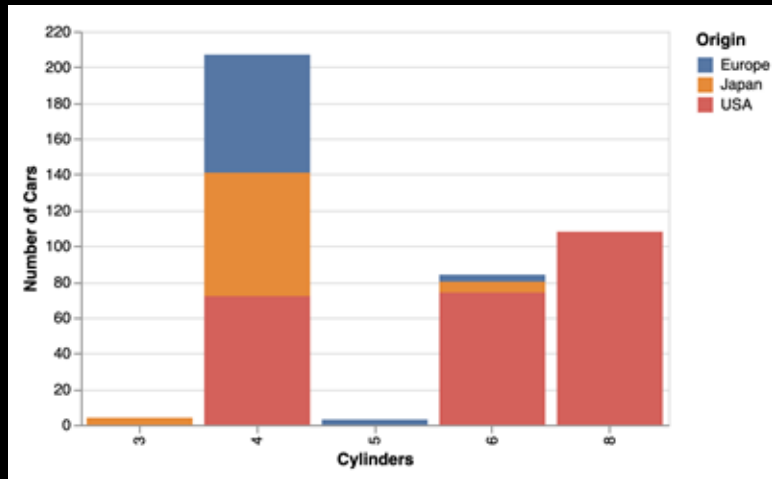
Encode (“Show me a different visual representation”)



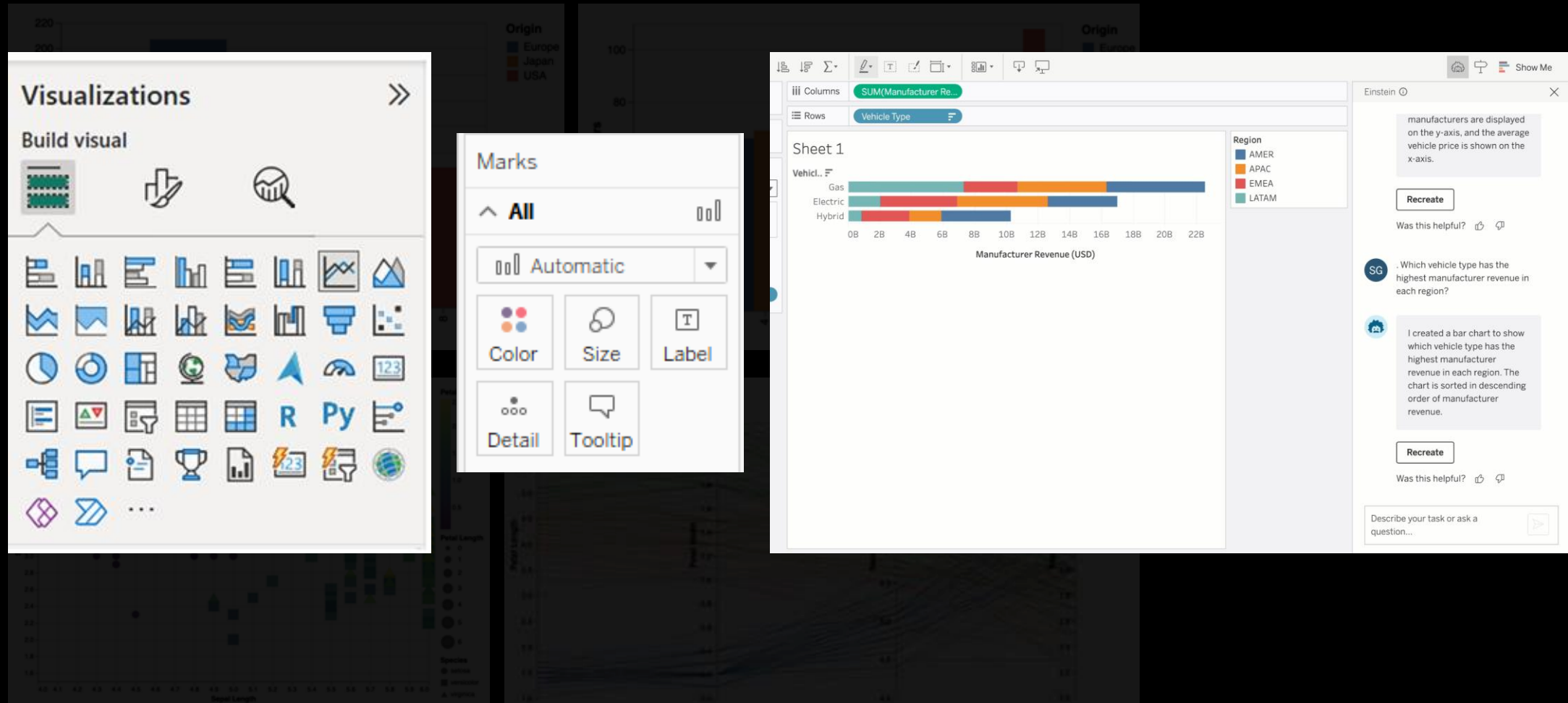
Encode (“Show me a different visual representation”)



Encode (“Show me a different visual representation”)



Encode (“Show me a different visual representation”)



Select—mark something as interesting.

Abstract/Elaborate—show me more or less detail.

Explore—show me something else.

Connect—show me related items.

Reconfigure—show me a different arrangement.

Filter—show me something conditionally.

Encode—show me a different representation.

Experiences that can be afforded through interaction

- [LineUp](#)
- [Dust & Magnets](#)
- ...

TLDR: Direct Manipulation + Dynamic Querying

If I had to call out one “interaction paradigm” that you should think about while designing interfaces containing visualizations

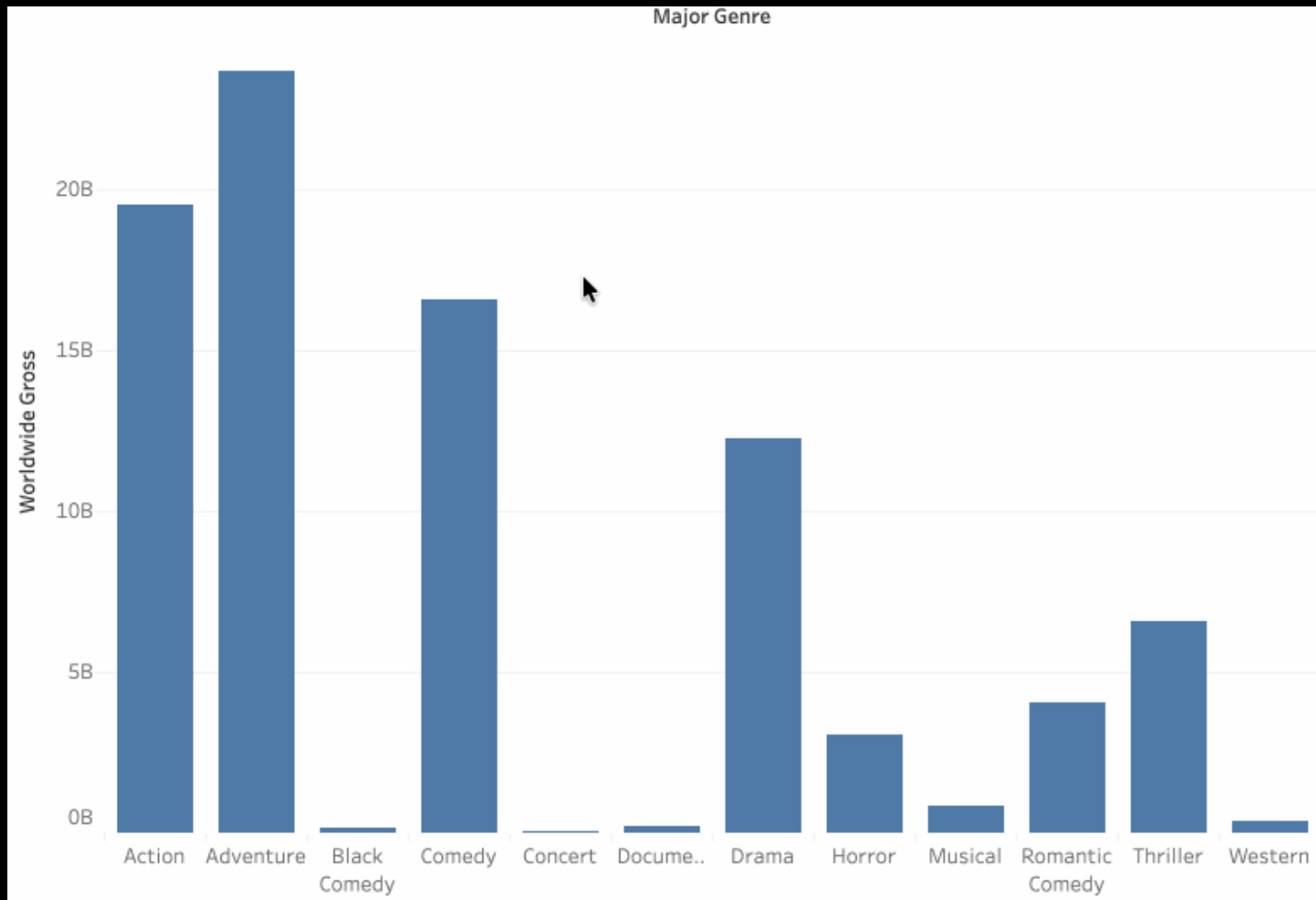
TLDR: Direct Manipulation + Dynamic Querying

The term *direct manipulation* was coined by [Shneiderman \(1974, 1982, 1983\)](#) to refer to systems having the following properties:

1. Continuous representation of the object of interest.
2. Physical actions or labeled button presses instead of complex syntax.
3. Rapid incremental reversible operations whose impact on the object of interest is immediately visible. ([Shneiderman, 1982, p. 251](#))

Hutchins et al., 1985

TLDR: Direct Manipulation + Dynamic Querying



- Sort icon directly on legend with immediate feedback
- Marks are always visible

TLDR: Direct Manipulation + Dynamic Querying

[1988](#)). An alternative database interface utilizing dynamic queries (Ahlberg, Williamson & Shneiderman, 1991) would:

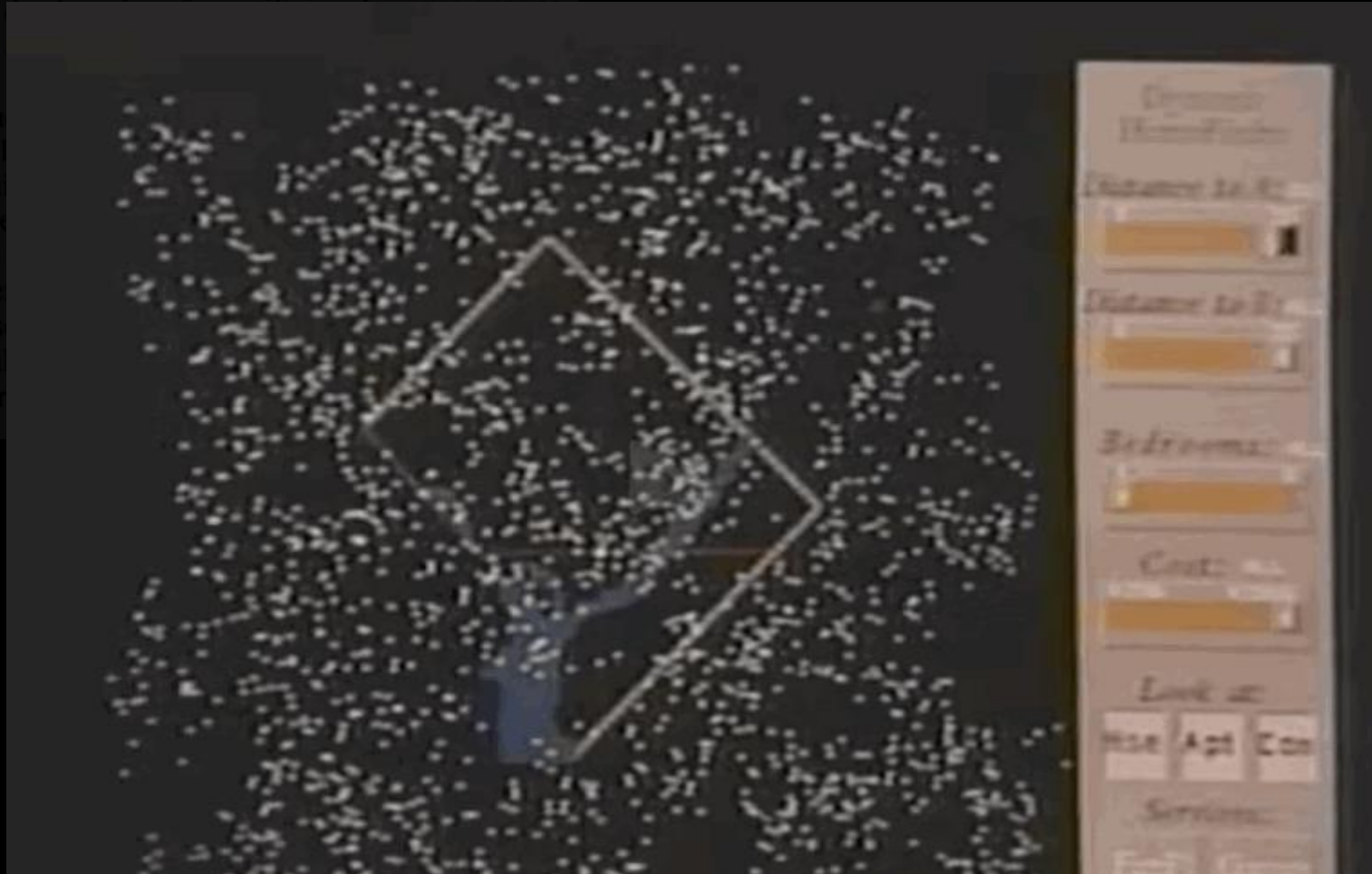
- represent the query graphically,
- provide visible limits on the query range,
- provide a graphical representation of the database and the query result,
- give immediate feedback of the result during every query adjustment, and
- allow novice users to begin working with little training, but still provide expert users with powerful features.

Dynamic queries for information exploration: An implementation and evaluation
Ahlberg et al., 1992

TLDR: Direct Manipulation + Dynamic Querying

1988). An alternative
(Ahlberg, Williamson

- represent the qu
- provide visible l
- provide a graph
- and the query r
- give immediate
- query adjustme
- allow novice us
- training, but sti
- powerful featur



Date From To

Store All

Product All

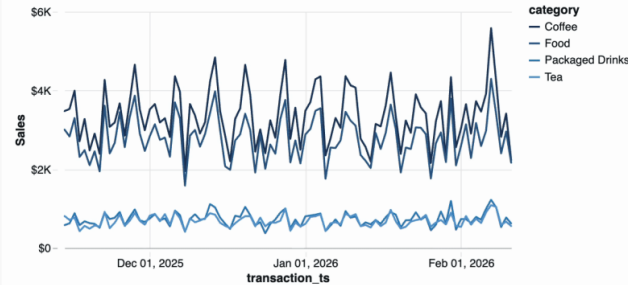
Daily Sales



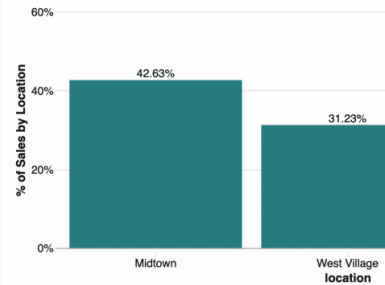
Sales

category	subcategory	product_name	Sales	Num Transactions	COGS	Gross Margin	Margin %
Coffee	Brewed Coffee	Affogato	\$13332.73	1960	3556.80	\$9775.93	73.32%
		Cortado	\$9367.44	1948	1675.35	\$7692.09	82.12%
		Drip Coffee (Large)	\$9363.45	2164	771.05	\$8592.40	91.77%
		Drip Coffee (Medium)	\$8244.79	2179	659.70	\$7585.09	92.00%
		Drip Coffee (Small)	\$7207.40	2193	554.50	\$6652.90	92.31%
		French Press	\$11601.91	1914	1353.10	\$10248.81	88.34%
		Macchiato	\$8479.80	1869	1413.00	\$7066.80	83.34%
		Pour Over	\$8780.09	1589	957.60	\$7822.49	89.09%
		Red Eye	\$9855.13	1953	788.40	\$9066.73	92.00%

Sales by Category



Distribution of Sales by Location



Product Usage and ROI

Action User Department
Select... Select... Select...

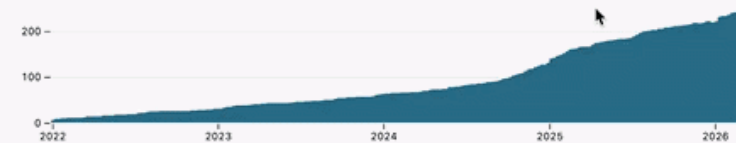
Unique Users

249

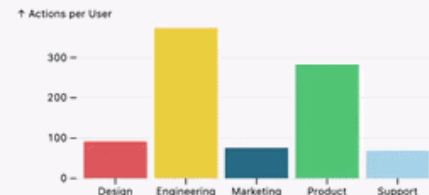
Avg. Session Duration (s)

79.90

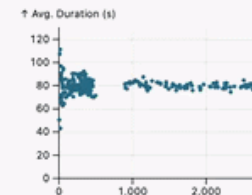
Daily Active Users



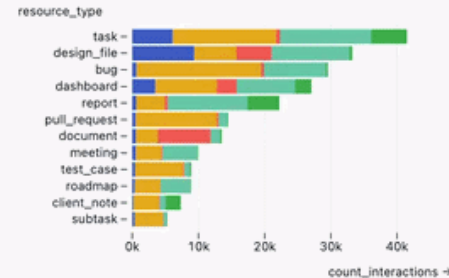
Department Profile



Users by Activity



Which resource types receive the most interactions, and how does that vary by department?



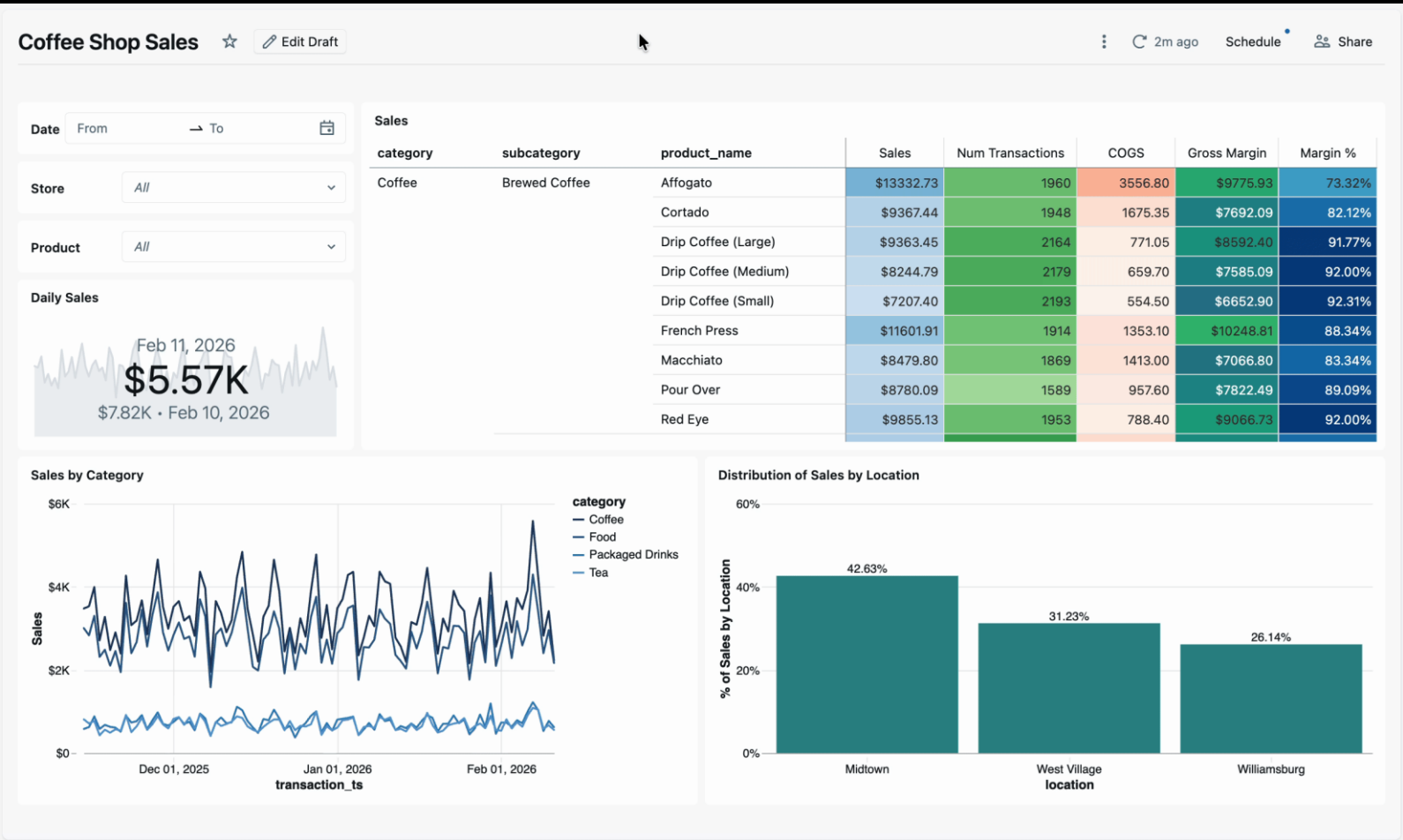
user_department
Design Engineering Marketing Product Support

Overall, the highest interaction volumes are concentrated in Engineering and Product. Engineering's most-interacted resource types are bugs (18,832), tasks (15,666), pull requests (12,277), dashboards (9,314), and test cases (7,313). Product's most-interacted types are tasks (13,704), reports (12,058), design files (11,766), bugs (9,332), dashboards (8,802), plus meetings (5,267) and roadmaps (4,571). Design activity is led by design files (9,342) and tasks (6,053). Marketing is led by documents (7,986) and design files (5,354). Support is led by tasks (5,463) and reports (4,833).

Query

Ask a question about the data

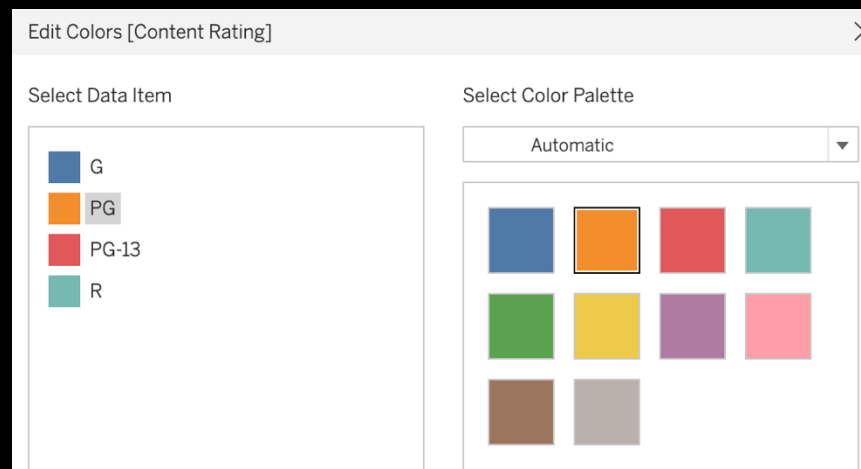




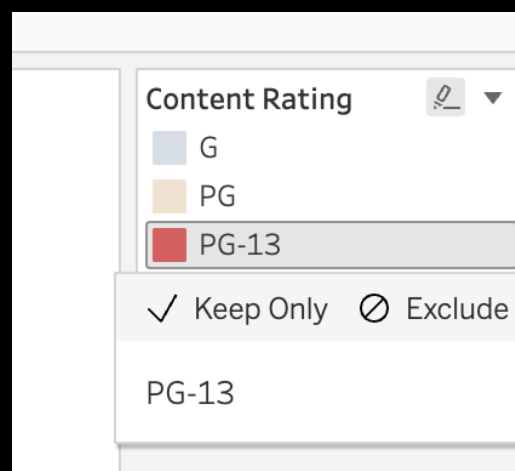
Users can **select** marks to see details and specify a filter. Charts across the dashboard are **connected** to support cross-filtering. Level of detail on charts like Pivot tables can be adjusted to **abstract/elaborate** data.

Now that we have chat/language input, can't people just ask/say what they want?

*“Switch color from orange to purple”/
“Color PG purple”*



“Just show PG-13 movies”



*“Show profitable PG-13 movies that were
made with a budget of under 100M”*

Maybe not? Or perhaps some combination of NL with an interactive widget (e.g., for budget and content rating)

*“Switch color from
“Color PG purple*

Almost never a single answer

- Who is the target user? What is their expertise level?
- What are the core tasks? How frequently are these performed?
- Are there latency, computation, or device constraints?

Point is to ask these questions, consider examples you’ve seen today and design/argue for the right choice.

*“Show profitable PG-13 movies that were
made with a budget of under 100M”*

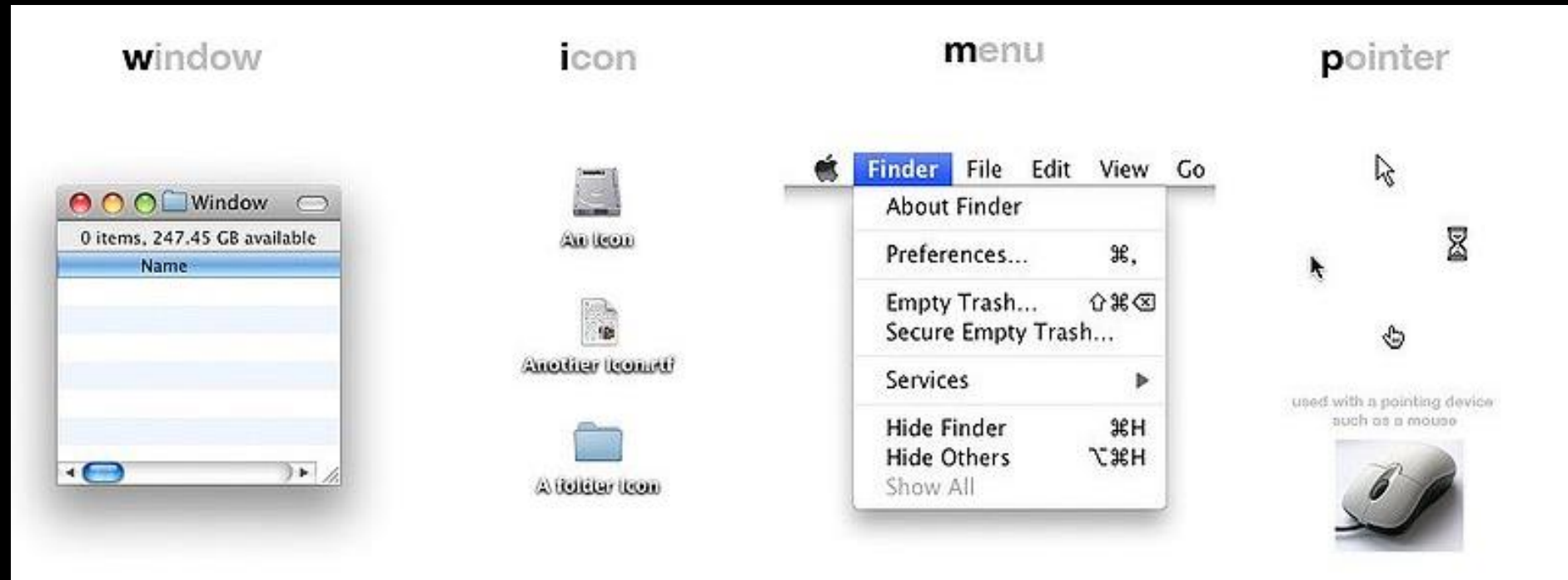
Maybe not? Or perhaps some combination of NL with an interactive widget (e.g., for budget and content rating)

Post-WIMP

A whole topic of its own. Goal here is to introduce the idea and share resources to spike interest.

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A whole topic of its own. Goal here is to introduce the idea and share resources to spike interest.







Post-WIMP



Touch:

- Tablets:
 - [Scatterplots](#)
 - [MCV](#)
 - [Kinetica](#)
 - [VisTiles](#)
- Tabletops/large displays:
 - [Cambiera](#)
 - [VisWall](#)
 - [David + Goliath](#)

Multi-modal:

- [DataInk](#)
- [InChorus](#)
- [Orko](#)
- [Data@Hand](#)
- [DataBreeze](#)

AR/VR:

- [ImAxes](#)
- [Augmented Chironomia](#)

Dynamic Physicalization:

- [Zooids](#)

Recap + Takeaways

- **Interaction can help users specify and accomplish their intent---** invest time in understanding what users are trying to do and how your system can best support that.
- **Think about directness and fluidity** of the user experience when designing interactions in data and visualization systems.
- The intent-focused interaction categories can be **useful for testing and prompting during AI-assisted system development** (more on this in the next session)

Acknowledgements

- John Stasko, Alex Endert, Cindy Xiong, and Songwen Hu. Georgia Tech
- Arvind Satyanarayan. Massachusetts Institute of Technology
- Emily Wall. Emory University

Thank you

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